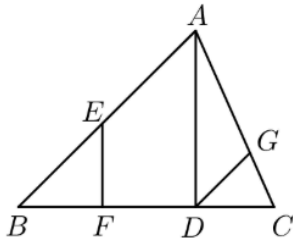


L1- L10 Class & HW

L1
Q10

As shown in the figure below, $\overline{AD} \perp \overline{BC}$, $\overline{EF} \perp \overline{BC}$ at D, F , respectively. $m\angle AEF + m\angle ADG = 180^\circ$, prove: $\overline{AB} \parallel \overline{GD}$



Answer

Since $\overline{AD} \perp \overline{BC}$ and $\overline{EF} \perp \overline{BC}$, we have $m\angle EFD = m\angle ADC = 90^\circ$, thus $\overline{AD} \parallel \overline{EF}$,
 $m\angle BEF = m\angle BAD$.

Since $m\angle BEF + m\angle AEF = 180^\circ$, we have $m\angle BAD + m\angle AEF = 180^\circ$.

Since we also have $m\angle AEF + m\angle ADG = 180^\circ$, thus we can get $m\angle BAD = m\angle ADG$, therefore
 $\overline{AB} \parallel \overline{GD}$ and $m\angle GDC = m\angle B$.

L2

Math Exploration 1 Question 4 Page 20

In triangle ABC , let D and E be on $\overline{AB}$ and $\overline{BC}$, respectively, such that $m\angle BAE = m\angle ACD$. Let F be the intersection of segments $\overline{AE}$ and $\overline{CD}$, and suppose that $\triangle CFE$ is equilateral. What is $m\angle BAC$? (Adapted from 2010 AMC 10A Problem, Question #14)

- A. 60° B. 75° C. 90° D. 105° E. 120°

Math Exploration 1 Question 4 Page 20

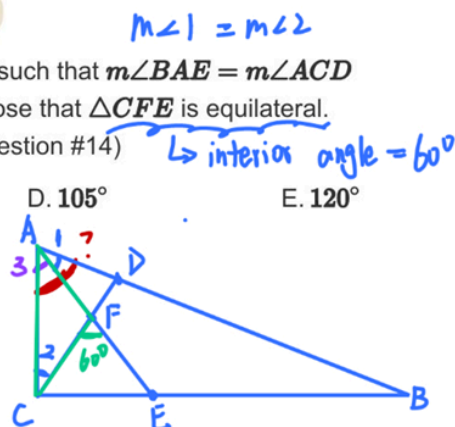
In triangle ABC , let D and E be on $\overline{AB}$ and $\overline{BC}$, respectively, such that $m\angle BAE = m\angle ACD$. Let F be the intersection of segments $\overline{AE}$ and $\overline{CD}$, and suppose that $\triangle CFE$ is equilateral. What is $m\angle BAC$? (Adapted from 2010 AMC 10A Problem, Question #14)

✓ A. 60° B. 75° C. 90° D. 105° E. 120°

$m\angle 3 + m\angle 2 = 60^\circ$

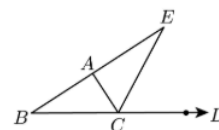
↓

$m\angle 3 + m\angle 1 = 60^\circ = m\angle BAC$



Q2

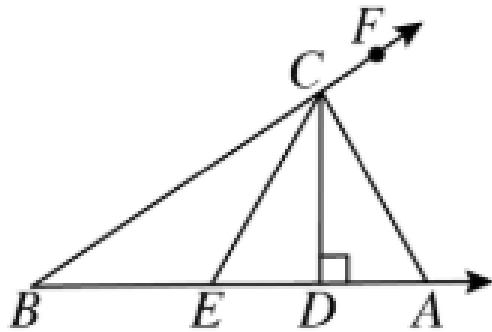
As shown in the figure below, $\overline{CE}$ is the angle bisector of $\angle ACD$ and $\overline{CE}$ intersects line BA at E , $m\angle B = 32^\circ$, $m\angle E = 26^\circ$, then $m\angle BAC$ is



Ans 84

Q6

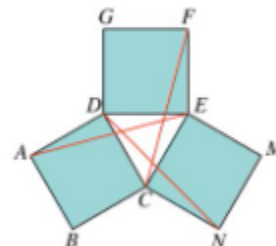
As shown in the figure below, in $\triangle ABC$, $\overline{CD}$ and $\overline{CE}$ are altitude and angle bisector of $\triangle ABC$, $m\angle BAC = 70^\circ$, $m\angle B = 40^\circ$. $m\angle DCE =$ $^\circ$.



Ans 15

Math Exploration 3

The diagram shows three congruent squares and an equilateral triangle. Prove that $AE = FC = ND$.

**Math Exploration 3**

The diagram shows three congruent squares and an equilateral triangle. Prove that $AE = FC = ND$.

In 3 congruent squares. $AD = FE = NC$
 $DE = EC = CD$

In equilateral $\triangle CED$, $m\angle CDE = m\angle DEC = m\angle DCE$
 $= 60^\circ$

Thus, $m\angle ADE = m\angle FEC = m\angle NCD = 90^\circ + 60^\circ = 150^\circ$.

In $\triangle ADE$, $\triangle FEC$, $\triangle NCD$



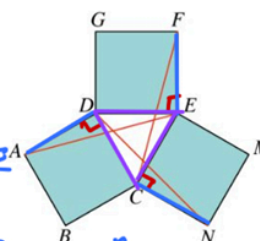
$$\begin{cases} AD = FE = NC \end{cases}$$

$$\begin{cases} m\angle ADE = m\angle FEC = m\angle NCD \end{cases}$$

$$\begin{cases} DE = EC = CD \end{cases}$$

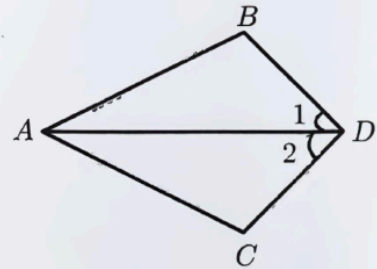
$$\Rightarrow \triangle ADE \cong \triangle FEC \cong \triangle NCD$$

$$\Rightarrow AE = FC = ND$$



Q2

- As shown in the figure below, $m\angle 1 = m\angle 2$, which of the following conditions, if added, still cannot deduce $\triangle ABD \cong \triangle ACD$?

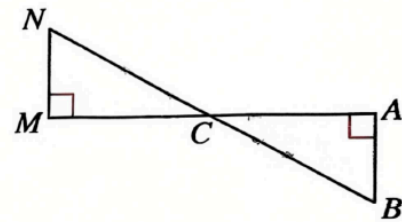


- A. $BD = CD$
C. $m\angle B = m\angle C$

- B. $AB = AC$
D. $m\angle BAD = m\angle CAD$

Ans B

- 7 Given that $m\angle M = m\angle A$; $NC = BC$, prove that $NM = BA$.

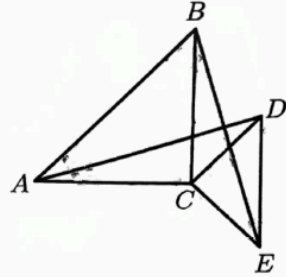


Answer

Plan for proof:

By AAS Theorem, $\triangle NMC \cong \triangle BAC$. Thus $NM = BA$.

- 9 As shown in the figure below, $\triangle ABC$ and $\triangle ECD$ are isosceles right triangles. $m\angle ACB = m\angle DCE = 90^\circ$, connect $\overline{AD}$ and $\overline{BE}$. Prove that $\triangle ACD \cong \triangle BCE$.



Answer

$AC = BC, CD = CE, m\angle ACB = m\angle DCE = 90^\circ$,
thus $m\angle ACD = m\angle BCE$.

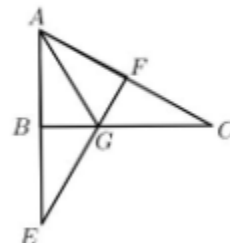
In $\triangle ACD$ and $\triangle BCE$,

$$\begin{cases} AC = BC \\ m\angle ACD = m\angle BCE, \\ CD = CE \end{cases}$$

so $\triangle ACD \cong \triangle BCE$ (SAS).

Math Exploration 3

2. In $\triangle ABC$, $m\angle ABC = 90^\circ$. Extend $\overline{AB}$ to E so that $AE = AC$. Create $\overline{EF} \perp \overline{AC}$ at F going through E such that $\overline{EF}$ intersects $\overline{BC}$ at G . Prove that $\overline{AG}$ bisects $\angle BAC$.



Math Exploration 3

2. In $\triangle ABC$, $m\angle ABC = 90^\circ$. Extend $\overline{AB}$ to E so that $AE = AC$. Create $\overline{EF} \perp \overline{AC}$ at F going through E such that $\overline{EF}$ intersects $\overline{BC}$ at G . Prove that $\overline{AG}$ bisects $\angle BAC$.

In $\triangle ABC$, $\triangle AFE$

$$\begin{cases} m\angle AFE = m\angle ABF = 90^\circ \\ m\angle EAC = m\angle EAC \text{ (common)} \\ AE = AC \end{cases}$$

$$\triangle ABC \cong \triangle AFE \text{ (AAS)}$$

$$\Rightarrow AB = AF$$

(corresponding sides)

$$\Rightarrow m\angle 1 = m\angle 2 ?$$

$$\Rightarrow \triangle AEG \cong \triangle ACG \text{ SS}$$

$$\triangle ABG \cong \triangle AFG \text{ AS}$$

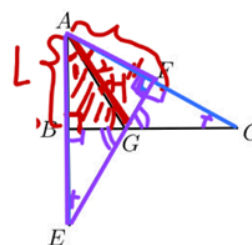
$$\text{In } \triangle ABG, \triangle AFG$$

$$\begin{cases} AG = AG \end{cases}$$

$$AB = AF$$

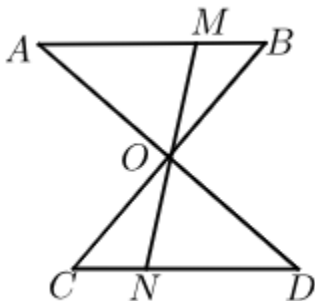
$$\text{Rt } \triangle ABG \cong \text{Rt } \triangle AFG \text{ (HL)} \Rightarrow m\angle 1 = m\angle 2$$

$\Rightarrow \overline{AG}$ bisects



Q3

As shown in the figure, $\overline{AD}$ and $\overline{BC}$ intersect at point O , and $OA = OD$, $OB = OC$. Create $\overline{MN}$ through point O that intersects $\overline{AB}$ at M and $\overline{CD}$ at N . Prove that $OM = ON$.



Paragraph proof:

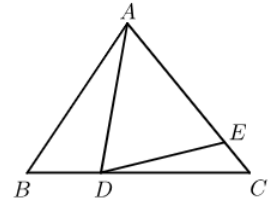
Since $OA = OD$, $m\angle AOB = m\angle DOC$, and $OB = OC$, we have $\triangle AOB \cong \triangle DOC$ (SAS), so $m\angle A = m\angle D$. Since $OA = OD$ and $m\angle AOM = m\angle DON$, we have $\triangle AOM \cong \triangle DON$ (ASA), so $OM = ON$.

Proof:

Statements	Reasons
1. $m\angle AOB = m\angle DOC$	1. Vertical angles are equal
2. $OA = OD, OB = OC$	2. Given
3. $\triangle AOB \cong \triangle DOC$	3. SAS postulate
4. $m\angle A = m\angle D$	4. Corresponding angles of congruent triangles are equal.
5. $m\angle AOM = m\angle DON$	5. Vertical angles are equal
6. $\triangle AOM \cong \triangle DON$	6. ASA postulate
7. $OM = ON$	7. Corresponding sides of congruent triangles are equal.

Q8

It is given that two base angles of an isosceles triangle are equal to each other. As shown in the diagram, then in $\triangle ABC$, $AB = AC$, $m\angle BAD = 30^\circ$, and $AE = AD$. Then $m\angle EDC = \underline{\hspace{1cm}}$.



- A) 15
- B) 10
- C) 5
- D) 20

Answer

A

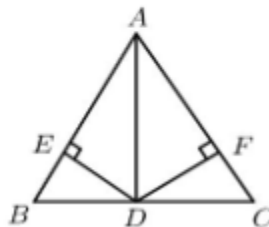
Solution

Let $m\angle EDC = \alpha$, $m\angle C = m\angle B = \beta$, then $m\angle AED = m\angle ADE = \alpha + \beta$.

From $m\angle B + m\angle BAD = m\angle ADC = m\angle ADE + m\angle EDC$, we have $\beta + 30^\circ = \alpha + \beta + \alpha$, which is equivalent to $\alpha = 15^\circ$.

Math Exploration 2

4. As shown in the figure, in $\triangle ABC$, D is the midpoint of $\overline{BC}$. $\overline{DE} \perp \overline{AB}$ and $\overline{DF} \perp \overline{AC}$ at point E and F , respectively. Given that $BE = CF$, prove that $\overline{AD}$ bisects $\angle BAC$.



Math Exploration 2

4. As shown in the figure, in $\triangle ABC$, D is the midpoint of $\overline{BC}$. $\overline{DE} \perp \overline{AB}$ and $\overline{DF} \perp \overline{AC}$ at point E and F , respectively. Given that $BE = CF$, prove that $\overline{AD}$ bisects $\angle BAC$.

Since D is midpoint of $\overline{BC}$, $\overline{BD} = \overline{CD}$

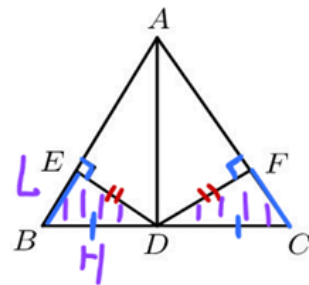
In $\triangle BDE$, $\triangle CDF$

$$\begin{cases} m\angle BED = m\angle CFD = 90^\circ \\ BD = CD \\ BE = CF \text{ (given)} \end{cases}$$

$$\triangle BDE \cong \triangle CDF \text{ (HL)}$$

$$\Rightarrow \overline{DE} = \overline{DF}$$

$$\Rightarrow \overline{AD} \text{ bisects } \angle BAC.$$



Math Exploration 3

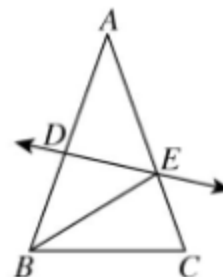
1. As shown in the figure below, in $\triangle ABC$, $m\angle ABC = m\angle C$, $AB = 8$, $\overleftrightarrow{DE}$ is the perpendicular bisector of $\overline{AB}$ and intersects $\overline{AB}$ and $\overline{AC}$ at D , E , respectively. The perimeter of $\triangle BEC$ is 13, $BC = \underline{\hspace{1cm}}$.

A. 5

B. 6

C. 7

D. 8



Math Exploration 3

1. As shown in the figure below, in $\triangle ABC$, $m\angle ABC = m\angle C$, $AB = 8$, $\overleftrightarrow{DE}$ is the perpendicular bisector of $\overline{AB}$ and intersects $\overline{AB}$ and $\overline{AC}$ at D , E , respectively. The perimeter of $\triangle BEC$ is 13, $BC = \underline{\hspace{1cm}}$.

A. 5

B. 6

C. 7

D. 8

Since $m\angle ABC = m\angle C$, $\triangle ABC$ is iso $\triangle$. $AB = AC = 8$

Since $\overleftrightarrow{DE}$ is perpendicular bisector, $AE = BE$

$$\text{Perimeter } \triangle BEC = BE + CE + BC = 13$$

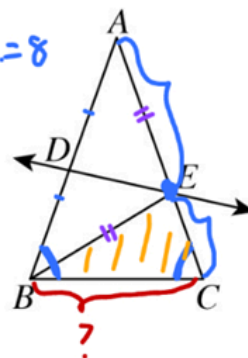
$$\downarrow$$

$$(AE + CE) + BC = 13$$

$\downarrow$

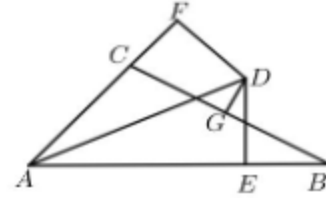
$$8 + BC = 13$$

$$BC = 5$$



Math Exploration 3

2. As shown in the figure below, $\overline{AD}$ is the angle bisector of $\angle BAC$ and $\overleftrightarrow{DG}$ is perpendicular bisector of $\overline{BC}$. $\overline{AD}$ and $\overleftrightarrow{DG}$ intersect at D . $\overline{DE} \perp \overline{AB}$ at E , $\overline{DF} \perp \overline{AC}$ at F , $AB = 6$, $AC = 3$, then $BE =$ _____. (A little hard, try to connect some points!)



Math Exploration 3

2. As shown in the figure below, $\overline{AD}$ is the angle bisector of $\angle BAC$ and $\overleftrightarrow{DG}$ is perpendicular bisector of $\overline{BC}$. $\overline{AD}$ and $\overleftrightarrow{DG}$ intersect at D . $\overline{DE} \perp \overline{AB}$ at E , $\overline{DF} \perp \overline{AC}$ at F , $AB = 6$, $AC = 3$, then $BE =$ 1.5. (A little hard, try to connect some points!)

Since $\overline{AD}$ is angle bisector, $\overline{DF} = \overline{DE}$

connect $\overline{CD}$, $\overline{BD}$,

since $\overleftrightarrow{DG}$ is perpendicular bisector, $\overline{CD} = \overline{BD}$

In $\triangle CDF$, $\triangle BED$

$$\begin{cases} \angle F = \angle BED = 90^\circ \\ \overline{CD} = \overline{BD} \\ \overline{DF} = \overline{DE} \end{cases}$$

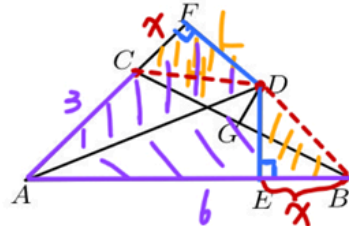
$\triangle CDF \cong \triangle BED$ (HL)

$$\Rightarrow \overline{EB} = \overline{FC}$$

$$\Rightarrow \overline{AF} = 3 + x = \overline{AE}, \triangle ADF \cong \triangle ADE$$

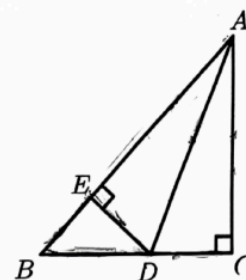
$$\overline{AB} = \overline{AE} + \overline{EB}$$

$$6 = 3 + x + x, x = 1.5$$





As shown in the figure, in $\triangle ABC$, $m\angle C = 90^\circ$, $\overline{AD}$ bisects $\angle BAC$. Draw $\overline{DE} \perp \overline{AB}$ at E through D . If $BC = 9$ and $BE = 3$, then the perimeter of $\triangle BDE$ is _____.



A. 15

B. 12

C. 9

D. 6

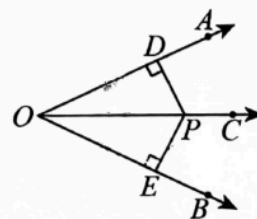
Answer

B

Solution

Since $\overline{AD}$ bisects $\angle BAC$ and $\overline{DE} \perp \overline{AB}$, $\overline{DC} \perp \overline{AC}$, we have $DE = DC$.
Then $DE + BE + DB = BC + BE = 12$.

- 8 As shown in the figure, $\overrightarrow{OC}$ is inside $\angle AOB$, P is an arbitrary point on the ray OC , $\overline{PD} \perp \overline{OA}$, $\overline{PE} \perp \overline{OB}$. Among the following conditions: ① $m\angle AOC = \frac{1}{2}m\angle AOB$; ② $PD = PE$; ③ $OD = OE$; ④ $m\angle DPO = m\angle EPO$, the number of conditions that can be used to deduce $\overrightarrow{OC}$ is the angle bisector of $\angle AOB$ is _____.



A. 1

B. 2

C. 3

D. 4

D

■ Solution

Since $m\angle AOC = \frac{1}{2}m\angle AOB$,

therefore, $\overrightarrow{OC}$ is the angle bisector of $\angle AOB$, which ① is correct.

Since $PD \perp OA$, $PE \perp OB$, and $PD = PE$,

therefore, $\overrightarrow{OC}$ is the angle bisector of $\angle AOB$, which ② is correct.

In right triangles POD and POE ,

$$\begin{cases} OD = OE \\ OP = OP \end{cases},$$

therefore, $\triangle POD$ is congruent to $\triangle POE$ (by HL),

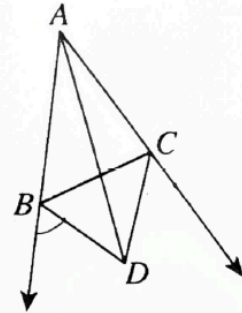
therefore, $m\angle AOC = m\angle BOC$,

therefore, $\overrightarrow{OC}$ is the angle bisector of $\angle AOB$, which ③ is correct.

Similarly, $\triangle POD$ is congruent to $\triangle POE$,

therefore, $m\angle AOC = m\angle BOC$,

As shown in the figure below, two exterior angle bisectors of $\triangle ABC$ intersect at D .
Prove: $\overline{AD}$ is the angle bisector of $\angle BAC$.



Answer

Paragraph proof:

Add $\overline{DE}$, $\overline{DF}$ and $\overline{DG}$ perpendicular to $\overline{AB}$, $\overline{BC}$ and $\overline{AC}$ at E , F , G , respectively.

Since $\overline{BD}$ bisects $\angle CBE$, $\overline{CD}$ bisects $\angle BCG$,

we have $DE = DF$, $DG = DF$,

then $DE = DG$,

thus point D is on the bisector of $\angle BAC$, $\overline{AD}$ is the angle bisector of $\angle BAC$.

Math Exploration 2

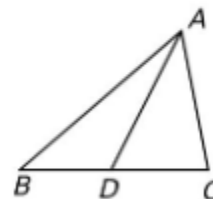
1. As shown in the figure below, in $\triangle ABC$, D is the median of $\overline{BC}$, $AC = 3$, $AD = 4$. Then the length of $\overline{AB}$ cannot be _____.

A. 5

B. 7

C. 8

D. 9



Math Exploration 2

1. As shown in the figure below, in $\triangle ABC$, D is the median of $\overline{BC}$, $AC = 3$, $AD = 4$. Then the length of $\overline{AB}$ cannot be _____.

A. 5

B. 7

C. 8

D. 9

Triangle inequality.

Extend AD to K , let $DK = AD$, then connect BK .

$$\begin{cases} BD = CD \\ \angle BDK = \angle CDA \\ KD = AD \end{cases}$$

$\triangle BDK \cong \triangle CDA$ (SAS)

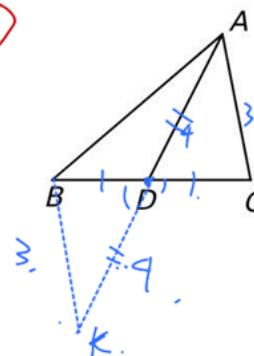
$$\Rightarrow BK = AC = 3$$

$$DK = AD = 4$$

$$AK = 8$$

$$8 - 3 < AB < 8 + 3$$

$$5 < AB < 11$$



Math Exploration 3

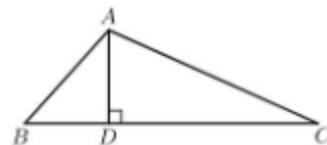
2. As shown in the figure below, in $\triangle ABC$, $m\angle BAC = 108^\circ$, $\overline{AD} \perp \overline{BC}$ at D , $AB + BD = CD$, then $m\angle C$ is _____.

A. 20°

B. 24°

C. 30°

D. 36°



Math Exploration 3

2. As shown in the figure below, in $\triangle ABC$, $m\angle BAC = 108^\circ$, $\overline{AD} \perp \overline{BC}$ at D , $AB + BD = CD$, then $m\angle C$ is _____.

A. 20°

B. 24°

C. 30°

D. 36°

Let point K on BC , $DK = BD$.
Connect AK .

Since D is the midpoint, $AD \perp BK$.

By perpendicular bisector thm.

$\Rightarrow AB = AK$.

$\triangle ABD \cong \triangle AKD$ (SAS)

$AB + BD = CD$

$AK + DK = CD$

$AK + DK = CD$

$AK = CK$

Let $c = x$, $\angle CAC = x$.

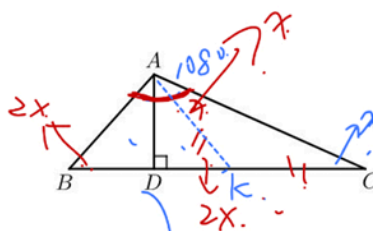
$108^\circ - x = 180^\circ - 4x$

$3x = 72^\circ$

$x = 24^\circ$

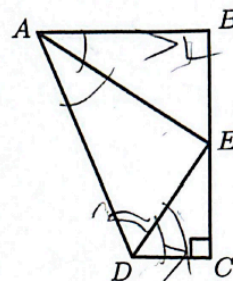
$\angle BAK = 108^\circ - x$

$= 180^\circ - 4x$



3

In the quadrilateral $ABCD$ shown in the figure, $\overline{AB} \parallel \overline{CD}$. If the angle bisector of $\angle DAB$, denoted as $\overline{AE}$, intersects $\overline{BC}$ at E , and $\overline{DE}$ bisects $\angle ADC$. Determine whether the following conclusion is true or false.



1. $CD + AB = BC$

2. E is the midpoint of $\overline{BC}$

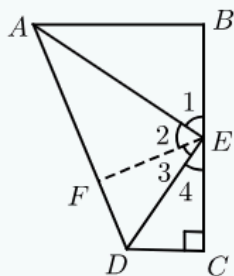
3. $m\angle AED = 90^\circ$ ✓ CP

4. the area of triangle $ADE = \frac{1}{2}$ the area of quadrilateral $ABCD$

Answer

FTTT

Solution



Let $\overline{EF} \perp \overline{AD}$ at F , as shown in the figure.

Since $\overline{AE}$ bisects $\angle DAB$, $\overline{EB} \perp \overline{AB}$, and $\overline{EF} \perp \overline{AD}$, we have $EF = EB$.

Since $\overline{AB} \parallel \overline{CD}$, we have $\overline{BC} \perp \overline{DC}$.

Since $\overline{DE}$ bisects $\angle ADC$, $\overline{BC} \perp \overline{DC}$, and $\overline{EF} \perp \overline{AD}$, we have $EF = EC$.

Therefore, $BE = EF = EC$, so statement ② is correct.

Since $\overline{AE}$ bisects $\angle DAB$ and $\overline{DE}$ bisects $\angle ADC$, we have $m\angle 1 = m\angle 2$ and $m\angle 3 = m\angle 4$.

Therefore, $m\angle AED = 90^\circ$, so statement ③ is correct.

In right triangles ABE and AFE , we have $\begin{cases} AE = AE \\ EF = EB \end{cases}$.

Therefore, $\text{Rt}\triangle ABE$ is congruent to $\text{Rt}\triangle AFE$ (by HL congruence).

Therefore, $AB = AF$.

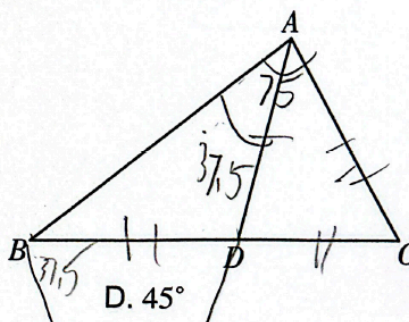
Similarly, we can prove that $\text{Rt}\triangle DFE$ is congruent to $\text{Rt}\triangle DCE$.

Therefore, $DF = DC$.

Therefore, $AD = AF + DF = AB + CD$, so statement ① is incorrect.

Since right triangle ABE is congruent to right triangle AFE and $\text{Rt}\triangle DEF$ is congruent to $\text{Rt}\triangle DEC$, we have the area of triangle $ADE = \frac{1}{2}$ the area of quadrilateral $ABCD$, so statement ④ is correct.

- 

B. 35° $C\checkmark 37.5^\circ$

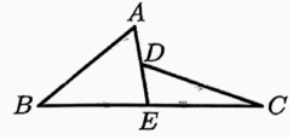
B

If we intersect AB on E such that $AE = AC$, then $\triangle ADE \cong \triangle ACD$.

Let $m\angle B = x^\circ$, then $m\angle BDE = x^\circ$, then $\angle AED = 2x^\circ$. Therefore, $m\angle C = 2x^\circ$. Thus we have $x^\circ + 2x^\circ + 75^\circ = 180^\circ$.

Solving this equation, we get $x = 35$.

- 5 As shown below, $\angle A = 60^\circ$, point E is a midpoint on line BC . D is a point on line AE . $AB = CD = 3$. Find the value of $AE + DE = \underline{\hspace{2cm}}$.



Answer

3

Solution

Extend DE to point F , making $EF = DE$. Connect BF ,

In $\triangle DEC$ and $\triangle FEB$,

$$\begin{cases} DE = FE \\ \angle DEC = \angle FEB, \\ CE = BE \end{cases}$$

$\therefore \triangle DEC \cong \triangle FEB$ (SAS),

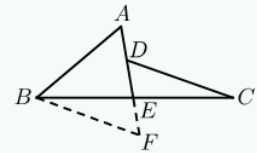
$\therefore BF = CD = 3$,

$\therefore AB = BF = 3$,

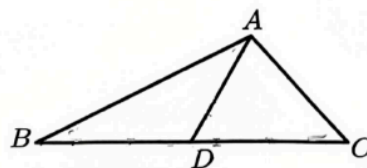
$\because \angle A = 60^\circ$,

$\therefore \triangle ABF$ is equilateral triangle.

$\therefore AF = AE + EF = AE + DE = 3$.



- 7 In $\triangle ABC$, $BD = CD$, $\angle DAC = \angle ABC + \angle ACB$. Given that $AD = 3$, find the length of $AB = \underline{\hspace{2cm}}$.

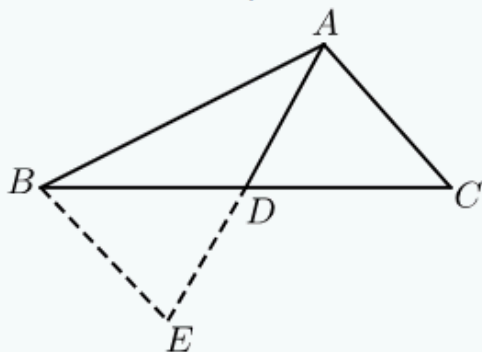


Answer

6

Solution

Extend AD to point E , making $DE = AD$, connect BE ,



$$\text{In } \triangle ADC \text{ and } \triangle EDB, \begin{cases} AD = ED \\ \angle ADC = \angle EDB, \\ CD = BD \end{cases}$$

$$\therefore \triangle ADC \cong \triangle EDB \text{ (SAS),}$$

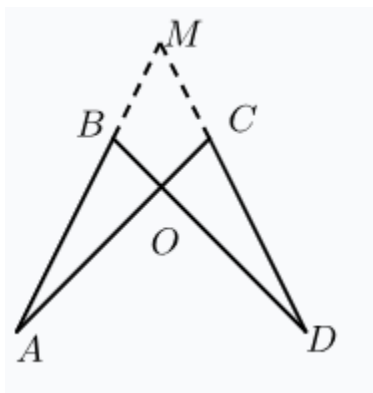
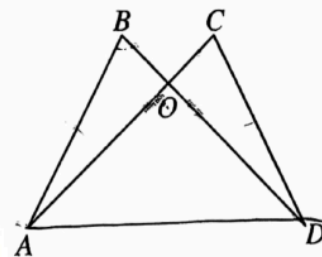
$$\therefore \angle DAC = \angle E, \angle C = \angle EBD,$$

$$\therefore \angle ABE = \angle ABC + \angle EBD = \angle ABC + \angle C, \angle DAC = \angle B + \angle C,$$

$$\therefore \angle ABE = \angle DAC = \angle E,$$

$$\therefore AB = AE = 2AD = 6.$$

- 10 As shown in the figure below, $AC = DB$, $m\angle B = m\angle C$, prove: $AB = CD$.



Extend $\overline{AB}$ and $\overline{DC}$ so that they intersect at M .

Since $m\angle ABD = m\angle DCA$,

we have $m\angle DBM = m\angle ACM$.

In $\triangle DBM$ and $\triangle ACM$,

$$\begin{cases} m\angle M = m\angle M \\ m\angle DBM = m\angle ACM, \\ DB = AC \end{cases}$$

then $\triangle DBM \cong \triangle ACM$ (AAS).

so $DM = AM$, $MB = MC$.

thus $AM - BM = DM - CM$,

hence $AB = CD$.

Math Exploration 2

2. How many right triangles have integer leg lengths a and b and a hypotenuse of length $b+1$, where $b < 100$? (2008 AMC 10B Problem, Question #15)

A. 6

B. 7

C. 8

D. 9

E. 10

Math Exploration 2

2. How many right triangles have integer leg lengths a and b and a hypotenuse of length $b+1$, where $b < 100$? (2008 AMC 10B Problem, Question #15)

A. 6

B. 7

C. 8

D. 9

E. 10

$$a^2 + b^2 = (b+1)^2$$

$$a^2 + \cancel{b^2} = \cancel{b^2} + 2b + 1$$

$$\underline{a^2 = 2b + 1}$$

a is an odd number.

$$2b + 1 < 201$$

$$\underline{a^2 < 201}$$

$$\underline{a = 3, 5, 7, 9, 11, 13}$$

Math Exploration 3

3. Point B is due east of point A . Point C is due north of point B . The distance between points A and C is $10\sqrt{2}$, and $\angle BAC = 45^\circ$. Point D is 20 meters due north of point C . The distance AD is between which two integers? (2012 AMC 10B Problem, Question #12)

A. 30 and 31

B. 31 and 32

C. 32 and 33

D. 33 and 34

E. 34 and 35

Math Exploration 3

3. Point B is due east of point A . Point C is due north of point B . The distance between points A and C is $10\sqrt{2}$, and $\angle BAC = 45^\circ$. Point D is 20 meters due north of point C . The distance AD is between which two integers? (2012 AMC 10B Problem, Question #12)

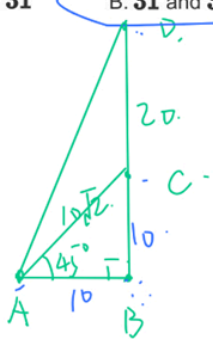
A. 30 and 31

B. 31 and 32

C. 32 and 33

D. 33 and 34

E. 34 and 35



$$AB = BC = 10$$

$$1:1:\sqrt{2}$$

$$DB = 20 + 10 = 30$$

$$\triangle ADB$$

$$AD = \sqrt{10^2 + 30^2}$$

$$= \sqrt{1000}$$

$$31^2 = 961$$

$$32^2 = 1024$$

$$31 < \sqrt{1000} < 32$$

2 A rectangular parking lot has a diagonal of 25 meters and an area of 168 square meters. In meters, what is the perimeter of the parking lot? (2011 AMC 10B Problem, Question #14)

A. 52

B. 58

C. 62

D. 68

E. 70

Answer

C

Solution

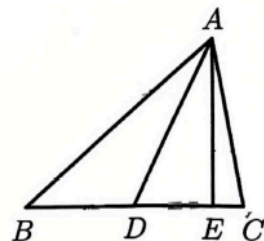
Let the sides of the rectangular parking lot be a and b . Then $a^2 + b^2 = 625$ and $ab = 168$. Add the two equations together, then factor.

$$a^2 + 2ab + b^2 = 625 + 168 \times 2$$

$$(a + b)^2 = 961$$

$$a + b = 31, \text{ The perimeter of a rectangle is } 2(a + b) = 2(31) = \boxed{(C) 62}.$$

- 8 As shown in the figure, in $\triangle ABC$, $\overline{AD}$ is the median from A to $\overline{BC}$, and $\overline{AE} \perp \overline{BC}$ at point E . If $AB = 12$, $BC = 10$, and $AC = 8$, then DE is _____.



Answer

4

Solution

Let $DE = x$.

By $\overline{AE} \perp \overline{BC}$ at point E :

$$AB^2 - BE^2 = AE^2 = AC^2 - CE^2.$$

Note that since $AB = 12$, $BC = 10$, $AC = 8$, and $BD = CD = 5$,

$$12^2 - (5 + x)^2 = 8^2 - (5 - x)^2.$$

Solve to get that $x = 4$, which means that $DE = 4$.

Math Exploration 2

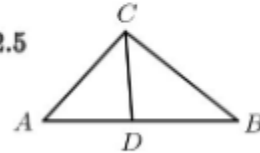
1. In $\triangle ABC$, $\overline{CD}$ is the angle bisector of $\angle ACB$, and $CA = 3$, $CB = 5$, $AB = 6$. The length of $\overline{AD}$ is _____.

A. 2

B. 3

C. $\frac{9}{4}$

D. 2.5



Math Exploration 2

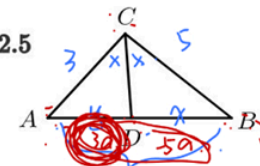
1. In $\triangle ABC$, $\overline{CD}$ is the angle bisector of $\angle ACB$, and $CA = 3$, $CB = 5$, $AB = 6$. The length of $\overline{AD}$ is _____.

A. 2

B. 3

C. $\frac{9}{4}$

D. 2.5



method ②

$$\Rightarrow \frac{AC}{CB} = \frac{AD}{DB}$$

$$\frac{3}{5} = \frac{3}{5}$$

③:5

$$\frac{3}{y} = \frac{5}{x}$$

$$\begin{cases} 3x = 5y \\ x + y = 6 \end{cases}$$

$$\begin{cases} x = \frac{9}{4} \\ y = \frac{15}{4} \end{cases}$$

$$6 \times \frac{3}{3+5} = 6 \times \frac{3}{8} = \frac{9}{4}$$

$$6 = 8a$$

$$a = \frac{6}{8} = \frac{3}{4}$$

$$3a = \frac{3}{4} \times 3 = \frac{9}{4}$$

Math Exploration 2

2. Trapezoid $ABCD$ has $\overline{AD} \parallel \overline{BC}$, $BD = 1$, $m\angle DBA = 23^\circ$, and $m\angle BDC = 46^\circ$. The ratio $BC : AD$ is $9 : 5$. What is CD ? (2009 AMC 12B Problem, Question #16)

A. $\frac{7}{9}$

B. $\frac{4}{5}$

C. $\frac{13}{15}$

D. $\frac{8}{9}$

E. $\frac{14}{15}$

Math Exploration 2

2. Trapezoid $ABCD$ has $\overline{AD} \parallel \overline{BC}$, $BD = 1$, $m\angle DBA = 23^\circ$, and $m\angle BDC = 46^\circ$. The ratio $BC : AD$ is $9 : 5$. What is CD ? (2009 AMC 12B Problem, Question #16)

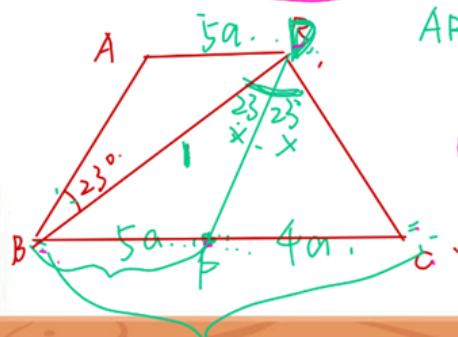
A. $\frac{7}{9}$

B. $\frac{4}{5}$

C. $\frac{13}{15}$

D. $\frac{8}{9}$

E. $\frac{14}{15}$



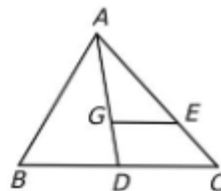
$AB \parallel DF$

$$\frac{DB}{BF} = \frac{DC}{CF}$$

$$\frac{1}{5a} \times \frac{DC}{4a} \Rightarrow DC = \frac{4}{5}$$

Math Exploration 3

1. As shown in the figure below, point G is the centroid of triangle ABC , and $\overline{GE}$ is parallel to $\overline{BC}$. If $BC = 12$, then $GE = \underline{\hspace{2cm}}$.



Math Exploration 3

6 equal parts

1. As shown in the figure below, point G is the centroid of triangle ABC , and $\overline{GE}$ is parallel to $\overline{BC}$. If $BC = 12$, then $GE = \underline{4}$.

G is centroid means D is the midpoint of BC

since G is on AD .

$$BD = CD = 6$$

Do the median of other 2 sides.

Try to get the ratio of $\frac{AG}{GD}$

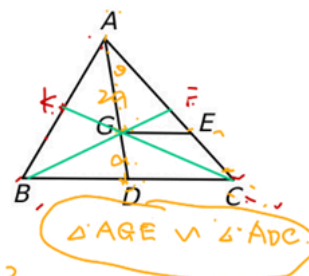
$$\text{Area of } \triangle ABG : \triangle BGD = 2 : 1$$

$$\Rightarrow AG : GD = 2 : 1$$

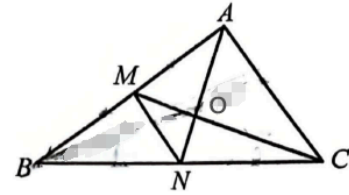
$$AG : AD = 2 : 3$$

$$\frac{AG}{AD} = \frac{2}{3}$$

$$\frac{GE}{DC} = \frac{4}{6}$$



3. As shown in the figure below, O is the centroid of triangle ABC . $\overline{AN}$ and $\overline{CM}$ intersect at point O . The area of triangle MON is 1. Find the area of triangle ABC .



Math Exploration 3

3. As shown in the figure below, O is the centroid of triangle ABC . $\overline{AN}$ and $\overline{CM}$ intersect at point O . The area of triangle MON is 1. Find the area of triangle ABC .

$$\frac{S}{S} = \frac{2}{1}, \quad \frac{A}{A} = \left(\frac{S}{S}\right)^2 = \frac{4}{1}$$

Area of $\triangle MON = 1$.

$$AO : ON = 2 : 1$$

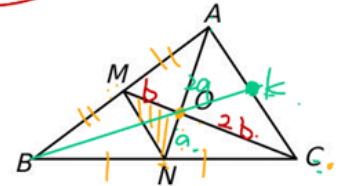
$$CO : MO = 2 : 1$$

$$\frac{CO}{MO} = \frac{AO}{NO} = \frac{2}{1}$$

$$\frac{A_{AOC}}{A_{MON}} = \frac{4}{1}$$

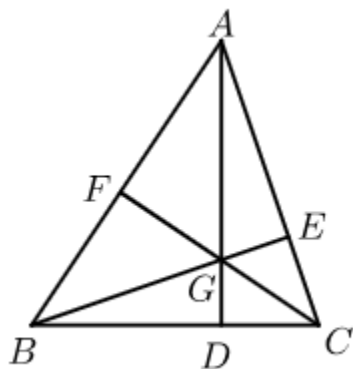
$$\Rightarrow \triangle AOC's \text{ Area} = 4$$

$$ABC = 4 \times 3 = 12$$



Q2

$\overline{AD} \perp \overline{BC}$ at point D , $\overline{BE} \perp \overline{AC}$ at point E , $\overline{CF} \perp \overline{AB}$ at point F , and $\overline{AD}$, $\overline{BE}$, $\overline{CF}$ intersect at point G .



(1) Point G is the orthocenter of triangle _____ .

(2) Point A is the orthocenter of triangle _____ .

(1) ABC or ACB or BCA or BAC or CAB or CBA

(2) BCG or BGC or CBG or CGB or GBC or GCB

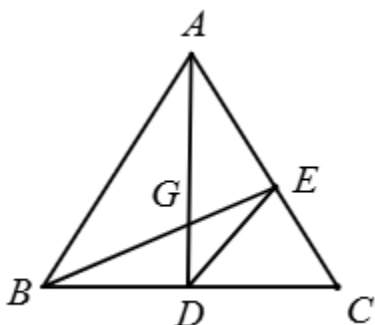
Solution

(1) Since $\overline{AD} \perp \overline{BC}$ at point D , $\overline{BE} \perp \overline{AC}$ at point E , and $\overline{CF} \perp \overline{AB}$ at point F , $\overline{AD}$, $\overline{BE}$, and $\overline{CF}$ intersect at point G , therefore point G is the orthocenter of $\triangle ABC$, so the answer is $\triangle ABC$.

(2) Since $\overline{AD} \perp \overline{BC}$ at point D , $\overline{BE} \perp \overline{AC}$ at point E , and $\overline{CF} \perp \overline{AB}$ at point F , $\overline{AD}$, $\overline{BF}$, and $\overline{CE}$ intersect at point A , therefore point A is the orthocenter of $\triangle BCG$, so the answer is $\triangle BCG$.

Q3

In the figure, it is known that $\overline{AD}$ is the median of triangle ABC , G is the centroid of triangle ABC . Connect $\overline{BG}$ and extend it to intersect side $\overline{AC}$ at point E . Connect $\overline{DE}$. Then, the ratio of the area of triangle ABC to the area of triangle GED is



Answer

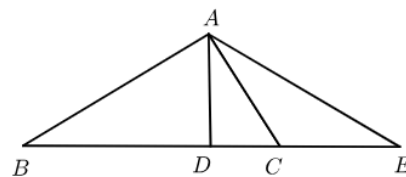
12 or 12 : 1

Solution

Since G is the centroid of triangle ABC , point D is the midpoint of $\overline{BC}$, and point E is the midpoint of $\overline{AC}$. Therefore, $\overline{BE}$ and $\overline{AD}$ are the medians of triangle ABC , and $\overline{DE}$ is the median of triangle ADC . Thus, $\overline{DE}$ is parallel to $\overline{AB}$, and $DE = \frac{1}{2}AB$. Therefore, $\triangle GDE$ is similar to triangle GAB , so $\frac{GE}{GB} = \frac{DE}{AB} = \frac{1}{2}$. Therefore, $BG = 2GE$, which means $BE = 3GE$. Using the formula for the area of a triangle, we have the area of triangle $BDE = 3$ the area of triangle GDE . Since D is the midpoint of $\overline{BC}$, the area of triangle $BCE = 2$ the area of triangle $BDE = 6$ the area of triangle GDE . Since E is the midpoint of $\overline{AC}$, the area of triangle $ABC = 2$ the area of triangle $BCE = 12$ triangle GDE . Therefore, the area of triangle ABC : the area of triangle $GED = 12$. Hence, the correct answer is 12.

Q4

In the figure, in right triangle ABC , $m\angle BAC = 90^\circ$, $\overline{AD} \perp \overline{BC}$ at D , construct $m\angle EAC = m\angle B$, and the extension of $\overline{AE}$ intersects with $\overline{BC}$ at E . If $AD = 3$ and $DE = 4$, then $DC =$, $AC =$.



Answer

$$\frac{3}{2} \text{ or } 1\frac{1}{2}; \frac{3\sqrt{5}}{2}$$

Solution

Since $AD = 3$ and $DE = 4$,
we have $AE = 5$.

Since $m\angle BAC = 90^\circ$ and $\overline{AD} \perp \overline{BC}$ at D ,
then $m\angle B = m\angle DAC$.

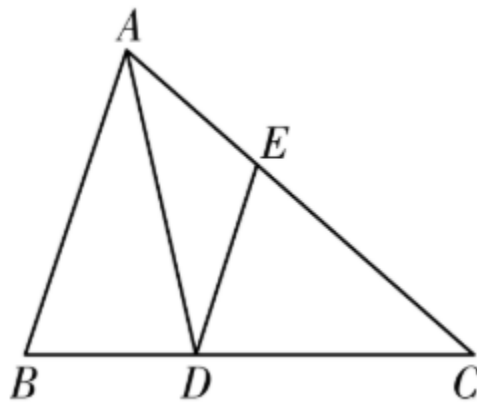
Since $m\angle EAC = m\angle B$,
we have $m\angle DAC = m\angle EAC$.

$$\text{thus } \frac{DC}{CE} = \frac{AD}{AE} = \frac{3}{5}.$$

$$\text{hence } CD = \frac{3}{2} \text{ and } AC = \frac{3\sqrt{5}}{2}.$$

Q8

Given the diagram, it is known that $\overline{AD}$ is the angle bisector of $\triangle ABC$, and $\overline{DE} \parallel \overline{AB}$. If $\frac{AE}{EC} = \frac{2}{3}$, find the value of $\frac{AC}{AB}$.



Solution

Since $\overline{AD}$ is the angle bisector of $\triangle ABC$,

we have $\frac{AC}{AB} = \frac{CD}{BD}$,

and $\overline{DE} \parallel \overline{AB}$,

therefore $\frac{CD}{BD} = \frac{CE}{AE} = \frac{3}{2}$.

Q10

In triangle ABC , $m\angle A = (x^2 + 6x + 12)^\circ$, $m\angle B = (3x + 8)^\circ$, $m\angle C = (18x)^\circ$, find the longest side of $\triangle ABC$.

Answer

$\overline{AB}$

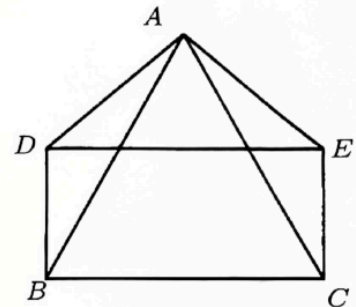
Solution

$$x^2 + 27x + 20 = 180, x = 5, m\angle C = 90^\circ.$$

So $\overline{AB}$ is the longest.

- As shown in the figure, it is given that $AB = AC$, $AD = AE$, $DE = BC$, and $m\angle BAD = m\angle CAE$.

Prove that quadrilateral $BCED$ is a rectangle.

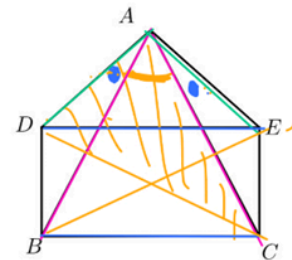


Math Exploration 4

2. As shown in the figure, it is given that $AB = AC$, $AD = AE$, $DE = BC$, and $m\angle BAD = m\angle CAE$.

Prove that quadrilateral $BCED$ is a rectangle.

- ① Prove $BCED$ is a parallelogram
 $\triangle BAD \cong \triangle CAE$ (SAS)
 $\Rightarrow DB = CE$
 Since $DE = BC$, $BCED$ is a parallelogram.
- ② Prove it's a rectangle



$$\angle DAB + \angle BAC = \angle EAC + \angle BAC$$

$$\angle DAC = \angle EAB$$

$$\begin{cases} AD = AE \\ \angle DAC = \angle EAB \end{cases}$$

$$\angle DAC = \angle EAB$$

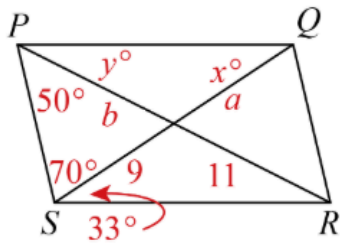
$$\triangle DAC \cong \triangle EAB \text{ (SAS)} \Rightarrow DC = BE$$

$$DC = BE$$

$\Rightarrow BCED$ is a rectangle

Q1

As shown in the figure below, $PQRS$ is a parallelogram. Find all the unknown angles and sides in it.

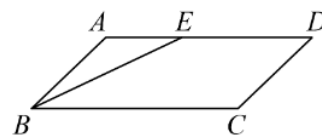


Answer

$$a = 9, b = 11, x = 33, y = 27$$

Q2

As shown in the figure, in parallelogram $ABCD$, E is the midpoint of side $\overline{AD}$. If $m\angle ABE = m\angle EBC$, $AB = 2$, then the perimeter of $ABCD$ is _____.



Answer

12

Since in parallelogram $ABCD$, $AB = 2$,
we have $CD = 2$.

Since $\overline{AD} \parallel \overline{BC}$,
we have $m\angle AEB = m\angle EBC$.

Note that $m\angle ABE = m\angle EBC$,
we have $m\angle ABE = m\angle AEB$,
so $AB = AE = 2$.

Since E is the midpoint of $\overline{AD}$,
we have $AD = BC = 2AE = 4$.

So the perimeter of the parallelogram $ABCD$ is 12.

- 8 Which of the following conditions can be used to prove that parallelogram $ABCD$ is a rectangle?

A. $CD = AB$ B. $m\angle A = m\angle C$ C. $AD = AB$ D. $AC = BD$

Answer

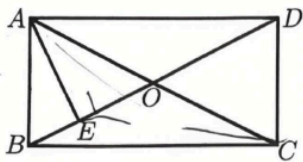
D

Solution

A and B cannot prove that $ABCD$ is a rectangle.

C can be used to determine that $ABCD$ is a rhombus, but that is not what the question was asking. Therefore D is the correct answer.

- 9 As shown, in rectangle $ABCD$, diagonals $\overline{AC}$ and $\overline{BD}$ intersect at O . $\overline{AE} \perp \overline{BD}$ at point E . $m\angle DAE : m\angle BAE = 3 : 1$. Then $m\angle EAC = \underline{\hspace{1cm}}^\circ$.



Answer

45

Solution

Since $m\angle DAB = m\angle DAE + m\angle BAE = 90^\circ$,
we have $m\angle DAE = 67.5^\circ$, $m\angle BAE = 22.5^\circ$.

Since $\overline{AE} \perp \overline{BD}$,

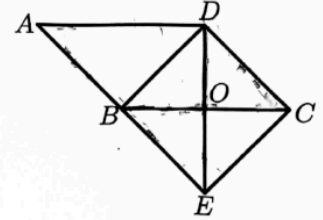
$m\angle BAE + m\angle ABE = 90^\circ$.

Since $m\angle BAE + m\angle DAE = 90^\circ$,
we have $m\angle ABE = m\angle DAE = 67.5^\circ$.

Lastly, since $AO = BO$,

we have $m\angle OAB = m\angle OBA = 67.5^\circ$,
therefore $m\angle EAC = 67.5^\circ - 22.5^\circ = 45^\circ$.

- 10 As shown in the figure below, in parallelogram $ABCD$, extend side $\overline{AB}$ to point E such that $BE = AB$. Draw $\overline{DE}$, $\overline{BD}$, and $\overline{EC}$. $\overline{DE}$ intersects with $\overline{BC}$ at point O . $m\angle BOD = 2m\angle A$. Prove that quadrilateral $BECD$ is a rectangle.



Solution

Since $ABCD$ is a parallelogram,

we have $AB = DC$, $\overline{AB} \parallel \overline{CD}$, and $m\angle A = m\angle BCD$.

Since $BE = AB$,

$\overline{BE} \parallel \overline{CD}$, $BE = DC$.

Therefore quadrilateral $BECD$ is a parallelogram.

So $OD = \frac{1}{2}DE$, $OC = \frac{1}{2}BC$.

Since $m\angle BOD = 2m\angle A$, and $m\angle BOD = m\angle OCD + m\angle ODC$,
we have $m\angle OCD = m\angle ODC$.

Therefore $OC = OD$,

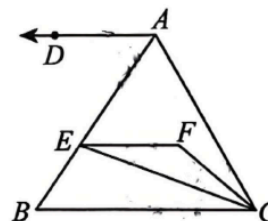
$DE = BC$,

so parallelogram $BECD$ is a rectangle.

Extention

L1

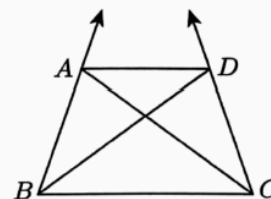
- 10 As shown in the figure below, $\overline{EF} \parallel \overline{AD}$, $\overline{AD} \parallel \overline{BC}$, $\overline{CE}$ bisects $\angle BCF$, $m\angle DAC = 3\angle BCF$, $m\angle ACF = 20^\circ$. Find $m\angle FEC$.



10. **【 Answer 】** Since $\overline{CE}$ bisects $\angle BCF$, suppose $m\angle BCE = m\angle ECF = \frac{1}{2}m\angle BCF = x$, thus $m\angle DAC = 3m\angle BCF = 6x$.
 Since $\overline{AD} \parallel \overline{EF}$ and $\overline{AD} \parallel \overline{BC}$, we have $\overline{EF} \parallel \overline{BC}$, therefore $m\angle DAC + m\angle ACB = 180^\circ$.
 Then we can get $6x + 2x + 20^\circ = 180^\circ$.
 Solve for x and we have $x = 20^\circ$.

L1 E

- 7 As shown in the figure below, given four conditions: ① $\overline{AC} = \overline{BD}$, ② $m\angle DAC = m\angle BCA$, ③ $m\angle ABD = m\angle CDB$, and ④ $m\angle ADB = m\angle CBD$, the conditions which can be used to prove $\overline{AD} \parallel \overline{BC}$ are _____.



A ①②

B. ③④

C ②④

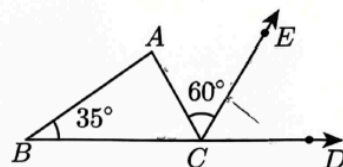
D. ①③④

1. **【 Answer 】** C

【 Solution 】 Since $m\angle DAC = m\angle BCA$, we have $\overline{AD} \parallel \overline{BC}$. (The alternate angles being equal implies that two line segments are parallel.)
 Since $m\angle ADB = m\angle CBD$, we have $\overline{AD} \parallel \overline{BC}$. (The alternate angles being equal implies that two line segments are parallel.)
 As such, statements ② and ④ can be used to prove $\overline{AD} \parallel \overline{BC}$.

L2 E

- 7 As shown in the figure below, $\overrightarrow{CE}$ is the bisector of $\angle ACD$. If $m\angle B = 35^\circ$, $m\angle ACE = 60^\circ$, then $m\angle A$ is _____.



A. 35°

B. 95°

C. 85°

D. 75°

1. **【 Answer 】** C

【 Solution 】 $m\angle ACE = 60^\circ$.

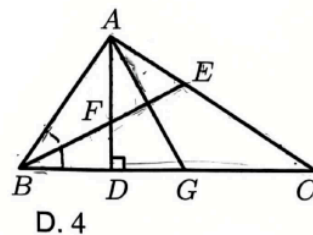
$$m\angle ACD = 2\angle ACE = 120^\circ,$$

Since $m\angle ACD = m\angle B + m\angle A$,

we have $m\angle A = m\angle ACD - m\angle B = 120^\circ - 35^\circ = 85^\circ$.

C.

- 10 As shown in the figure, in $\triangle ABC$, $m\angle BAC = 90^\circ$, $\overline{AD} \perp \overline{BC}$, the bisector of $\angle ABC$, $\overline{BE}$, intersects $\overline{AD}$ at point F , and $\overline{AG}$ bisects $\angle DAC$. Given the following statements: ① $m\angle BAD = m\angle C$; ② $m\angle AEF = m\angle AFE$; ③ $m\angle EBC = m\angle C$; ④ $\overline{AG} \perp \overline{EF}$. The number of correct statement(s) is _____.



A. 1

B. 2

C. 3

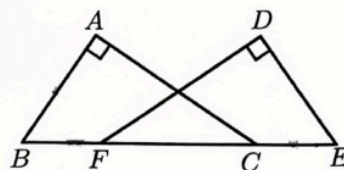
D. 4

10. **[Answer]** C

[Solution] Since $m\angle BAC = 90^\circ$, $\overline{AD} \perp \overline{BC}$,
we have $m\angle C + m\angle ABC = 90^\circ$, and $m\angle BAD + m\angle ABC = 90^\circ$,
so $m\angle BAD = m\angle C$, which means ① is correct.
Since $\overline{BE}$ is the bisector of $\angle ABC$,
we have $m\angle ABE = m\angle CBE$;
and since $m\angle ABE + m\angle AEF = 90^\circ$, $m\angle CBE + m\angle BFD = 90^\circ$,
we have $m\angle AEF = m\angle BFD$;
since $m\angle AFE = m\angle BFD$,
we have $m\angle AEF = m\angle AFE$, which means ② is correct.
Since $m\angle ABE = m\angle CBE$,
 $m\angle EBC = m\angle C$ is only true if $m\angle C = 30^\circ$. So ③ is incorrect.
Since $m\angle AEF = m\angle AFE$,
we have $AE = AF$.
Since $\overline{AG}$ bisects $\angle DAC$,
we have $\overline{AG} \perp \overline{EF}$. So ④ is correct .
The correct statements are ①②④ .
So C is the correct choice .

L3

9. As shown in the figure below, $m\angle A = m\angle D = 90^\circ$, $AB = DE$, $BF = EC$. Prove: $\text{Rt}\triangle ABC \cong \text{Rt}\triangle DEF$.



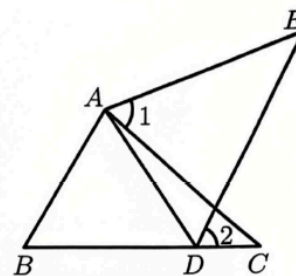
9. **【 Answer 】** Shown in the analysis.

【 Solution 】 Since $BF = EC$,
we have $BF + FC = FC + EC$, $BC = EF$,
thus $m\angle A = m\angle D = 90^\circ$,
so $\triangle ABC$ and $\triangle DEF$ are right triangles.
In $\text{Rt}\triangle ABC$ and $\text{Rt}\triangle DEF$,
$$\begin{cases} AB = DE \\ BC = EF \end{cases}$$

 $\text{Rt}\triangle ABC \cong \text{Rt}\triangle DEF$ (HL).

L3 E

9. As shown in the figure below, $m\angle 1 = m\angle 2$, in order to prove $\triangle ABC \cong \triangle ADE$, we can add condition _____.



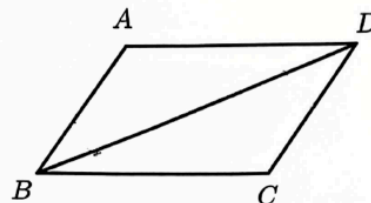
- A. $AB = AD$, $AC = AE$
C. $AB = DE$, $BC = AE$

- B. $AB = AD$, $BC = DE$
D. $AC = AE$, $BC = DE$

9. **【 Answer 】** D

【 Solution 】 N/A

10. As shown in the figure below, $AB = CD$, $\overline{AB} \parallel \overline{CD}$. Prove that $AD = BC$.



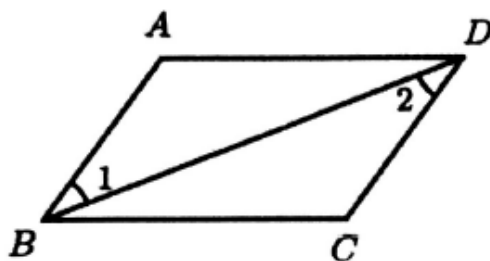
10. **【 Answer 】** $AD = BC$.

【 Solution 】 As shown in the figure below,
since $\overline{AB} \parallel \overline{CD}$,
we have $m\angle 1 = m\angle 2$.

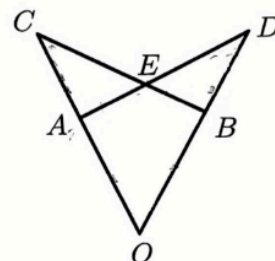
In $\triangle ABD$ and $\triangle CDB$,

$$\begin{cases} AB = CD \\ m\angle 1 = m\angle 2, \\ BD = DB \end{cases}$$

thus $\triangle ABD \cong \triangle CDB$ (SAS),
then $AD = CB$.



6. As shown in the diagram, $OA = OB$, $OC = OD$, $m\angle O = 60^\circ$, $m\angle C = 25^\circ$. The measure of $\angle BED$ is _____.



.. 70°
 C. 65°

B. 85°
 D. None of the above

6. **【 Answer 】** A

【 Solution 】 In $\triangle AOD$ and $\triangle BOC$,

$$\begin{cases} OA = OB \\ m\angle O = m\angle O, \\ OD = OC \end{cases}$$

so $\triangle AOD \cong \triangle BOC$ (SAS), which means $m\angle C = m\angle D$.

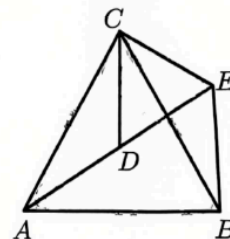
Since $m\angle C = 25^\circ$,
 we have $m\angle D = 25^\circ$.

Since $m\angle O = 60^\circ$, $m\angle C = 25^\circ$,
 we have $m\angle OBC = 95^\circ$.

Therefore $m\angle OBC = m\angle BED + m\angle D = 95^\circ$,
 so $m\angle BED = 70^\circ$.

So the correct choice is A.

- As shown in the figure below, $\triangle ACB$ and $\triangle DCE$ are equilateral triangles. D and E are on the same line. Connect $\overline{BE}$. $m\angle AEB$ is _____.



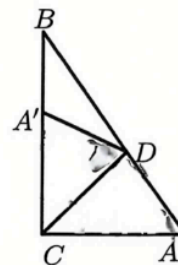
8. **【 Answer 】** Since $\triangle ACB$ and $\triangle DCE$ are equilateral triangles,
 we have $AC = BC$, $CD = CE$, $m\angle ACB = m\angle DCE = 60^\circ$, $m\angle CDE = m\angle CED = 60^\circ$,
 then $m\angle ACB - m\angle DCB = m\angle DCE - m\angle DCB$,
 thus $m\angle ACD = m\angle BCE$.
 In $\triangle ACD$ and $\triangle BCE$,

$$\begin{cases} AC = BC \\ m\angle ACD = m\angle BCE, \\ CD = CE \end{cases}$$

 so $\triangle ACD \cong \triangle BCE$,
 then $BE = AD$, $m\angle ADC = m\angle BEC$,
 thus $m\angle ADC = 180^\circ - 60^\circ = 120^\circ$,
 so $m\angle BEC = 120^\circ$,
 then we can get $m\angle AEB = m\angle BEC - m\angle CED = 120^\circ - 60^\circ = 60^\circ$.

L4 E

5. As shown in the figure, in $\text{Rt}\triangle ABC$, $m\angle ACB = 90^\circ$, $m\angle A = 55^\circ$. Fold the triangle such that A is on $\overline{CB}$ at the position marked as A' . The trace is $\overline{CD}$, then $m\angle A'DB =$ _____.



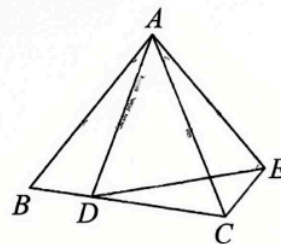
- A. 40° B. 30° C. 20° D. 10°

5. **【 Answer 】** C

【 Solution 】 From the question, it is implied that $\triangle A'DC \cong \triangle ADC$,
 so $m\angle DA'C = m\angle A = 55^\circ$, $m\angle A'CD = m\angle DCA = 45^\circ$.
 Therefore $m\angle A'DC = m\angle CDA = 80^\circ$.
 And thus $m\angle A'DB = 180^\circ - 80^\circ \times 2 = 20^\circ$.
 The correct answer is C.

 As shown in the figure below, point D is on $\overline{BC}$, $\triangle ABC$ and $\triangle ADE$ are equilateral triangles.

Prove: $m\angle ACE = m\angle BAC$.



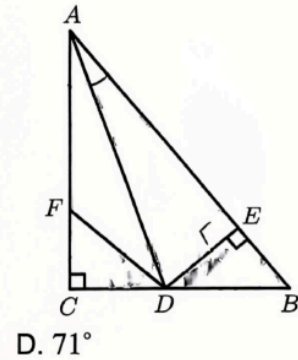
10. **【 Answer 】** $\because \triangle ABC$ and $\triangle ADE$ are equilateral triangles.
 $\therefore AB = AC, AD = AE, m\angle BAC = m\angle DAE = m\angle B = \angle 60^\circ$.
 $\therefore m\angle BAD + m\angle DAC = m\angle CAE + m\angle DAC, m\angle BAD = m\angle CAE$.
 In $\triangle ABD$ and $\triangle ACE$,

$$\begin{cases} AB = AC \\ m\angle BAD = m\angle CAE, \\ AD = AE \end{cases}$$

 $\therefore \triangle ABD \cong \triangle ACE (SAS)$,
 $\therefore m\angle ACE = m\angle B$,
 $\therefore m\angle ACE = m\angle BAC$.

L5

- 2 As shown in the figure below, in $\text{Rt}\triangle ABC$, $m\angle ACB = 90^\circ$, $m\angle B = 50^\circ$, D and F are points on $\overline{BC}$ and $\overline{AC}$, respectively. $\overline{DE} \perp \overline{AB}$ at E , $CF = BE$, $DF = DB$, then $m\angle ADE$ is _____.



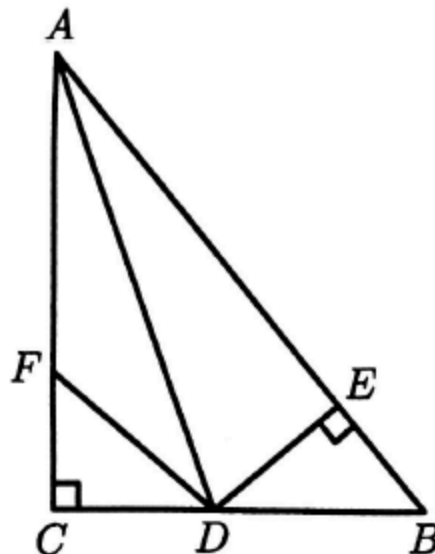
A. 40°

B. 50°

C. 70°

D. 71°

2. 【 Answer 】 C
【 Solution 】



$$\begin{cases} FC = BE \\ DF = DB \end{cases}$$

we have $\text{Rt}\triangle CDF \cong \text{Rt}\triangle EDB$ (HL),

thus $CD = DE$.

In $\text{Rt}\triangle ACD$ and $\text{Rt}\triangle AED$,

$$\begin{cases} CD = DE \\ AD = AD \end{cases}$$

so $\text{Rt}\triangle ACD \cong \text{Rt}\triangle AED$ (HL),

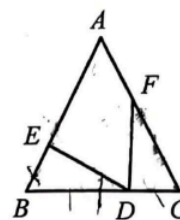
then $m\angle DAE = 20^\circ$,

thus $m\angle ADE = 70^\circ$.



As shown in the figure below, in $\triangle ABC$, $AB = AC$, $BE = CD$, $BD = CF$, then $m\angle EDF =$

_____.



A. $2m\angle A$

B. $90^\circ - 2m\angle A$

C. $90^\circ - m\angle A$

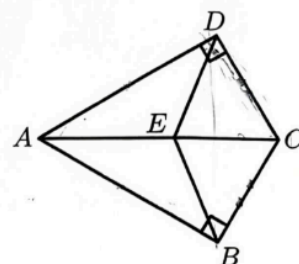
D. $90^\circ - \frac{1}{2}m\angle A$

5. **【 Answer 】** D

【 Solution 】 Since $AB = AC$,
 we have $m\angle B = m\angle C$,
 then $BE = CD$, $BD = CF$.
 So $\triangle BDE \cong \triangle CDF$ (SAS),
 thus $m\angle BED = m\angle CDF$.
 Since $m\angle B + m\angle BED + m\angle BDE = 180^\circ$,
 $m\angle EDF + m\angle CDF + m\angle BDE = 180^\circ$,
 we have $m\angle B = m\angle EDF$,
 thus $m\angle EDF = \frac{180^\circ - m\angle A}{2} = 90^\circ - \frac{1}{2}m\angle A$.

9. As shown in the figure below, given that $m\angle ABC = m\angle ADC = 90^\circ$, E is a point on $\overline{AC}$, $AB = AD$.

Prove: $EB = ED$.



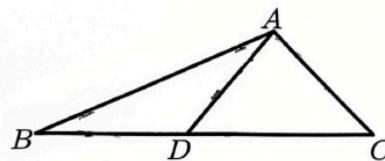
9. **【 Answer 】** Shown in the analysis.

【 Solution 】 Proof:

Statements	Reasons
1. $m\angle ABC = m\angle ADC = 90^\circ, AB = AD$	1. Given
2. $AC = AC$	2. Reflexive property
3. $\triangle ABC \cong \triangle ADC$	3. HL theorem
4. $m\angle DCE = m\angle BCE, DC = BC$	4. Corresponding sides/angles of congruent triangles are equal.
5. $CE = CE$	5. Reflexive property
6. $\triangle DCE \cong \triangle BCE$	6. SAS postulate
7. $EB = ED$	7. Corresponding sides of congruent triangles are equal.

L5 E

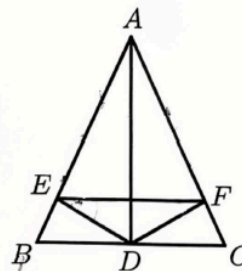
- 2 As shown in the figure, in $\triangle ABC$, D is on $\overline{BC}$ and $AC = AD = DB$. If $m\angle BAC = 102^\circ$, then $m\angle ADC = \underline{\hspace{1cm}}^\circ$.



2. **【 Answer 】** 52

【 Solution 】 Since $AC = AD = BD$,
 we have $m\angle BAD = m\angle DBA$ and $m\angle ADC = m\angle ACD$,
 so $m\angle BAD = \frac{1}{2}m\angle ADC$ and $m\angle DAC = 180^\circ - 2m\angle ADC$.
 Add the two equations together to obtain that
 $m\angle BAD + m\angle DAC = 180^\circ - \frac{3}{2}m\angle ADC = m\angle BAC = 102^\circ$.
 Solve to obtain $m\angle ADC = 52^\circ$.

- 3 As shown in the figure below, in $\triangle ABC$, $AB = AC$, $\overline{AD}$ bisects $\angle BAC$, $\overline{DE} \perp \overline{AB}$ at E and $\overline{DF} \perp \overline{AC}$ at F . Among the four following conditions: (1) $m\angle DEF = m\angle DFE$; (2) $AE = AF$; (3) $\overline{AD}$ bisects $\angle EDF$; (4) $\overline{EF}$ perpendicular bisects $\overline{AD}$, how many them are correct?



A. 1

B. 2

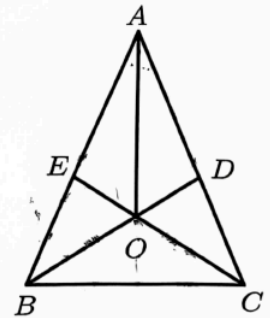
C. 3

D. 4

3. **【 Answer 】** C

【 Solution 】 Since $AB = AC$, $\overline{AD}$ bisects $\angle BAC$, $\overline{DE} \perp \overline{AB}$, $\overline{DF} \perp \overline{AC}$,
 we have $\triangle ABC$ is an isosceles triangle, $\overline{AD} \perp \overline{BC}$, $BD = CD$, $m\angle BED = m\angle DFC = 90^\circ$,
 thus $DE = DF$,
 so $\overline{AD}$ perpendicular bisect $\overline{EF}$,
 therefore (4) is incorrect.
 (1), $m\angle DEF = m\angle DFE$; (2) $AE = AF$; (3) AD bisects $\angle EDF$ are all correct.

- 6 As shown in the figure below, in $\triangle ABC$, $AB = AC$, $\overline{BD}$ and $\overline{CE}$ are angle bisectors of $\angle ABC$ and $\angle ACB$, respectively and they intersect at O . Connect $\overline{OA}$.



(2) Prove that line OA is perpendicular to $\overline{BC}$.

(2) In $\triangle AOB$ and $\triangle AOC$,

$$\begin{cases} AB = AC \\ AO = AO, \\ BO = CO \end{cases}$$

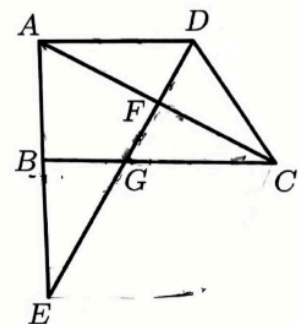
so $\triangle AOB \cong \triangle AOC$ (SSS),

thus $m\angle BAO = m\angle CAO$,

therefore $\overline{AO}$ bisects $\angle BAC$,

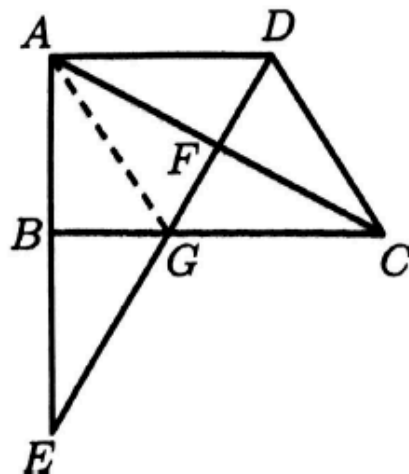
then $\overline{OA}$ is perpendicular to $\overline{BC}$.

- 8 As shown in the figure, in quadrilateral $ABCD$, $\overline{AD} \parallel \overline{BC}$, $m\angle ABC = 90^\circ$, $\overline{DE} \perp \overline{AC}$ at point F , intersecting $\overline{BC}$ at point G and $\overline{AE}$ at E . $AE = AC$. Prove that $BG = FG$. (Hint: Connect $\overline{AG}$)



8. 【 Answer 】 See analysis

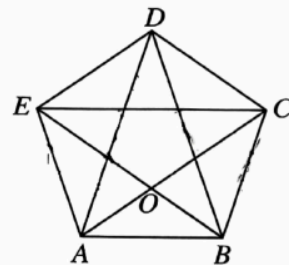
【 Solution 】 Since $m\angle ABC = 90^\circ$, $DE \perp AC$ at F ,
 we have $m\angle ABC = m\angle AFE$.
 Since $AC = AE$, we have $m\angle EAF = m\angle CAB$,
 we have $\triangle ABC \cong \triangle AFE$ (AAS),
 so $AB = AF$.
 Connect points to form AG ,
 since $AG = AG$, $AB = AF$,
 we have $\text{Rt}\triangle ABG \cong \text{Rt}\triangle AFG$ (HL).
 Therefore $BG = FG$.



10 As shown in the figure below, $\overline{AE} \parallel \overline{BD}$, $\overline{BC} \parallel \overline{AD}$, $AE = BC$ and $AD = BD$, prove that:

(1) $AC = BE$

(2) $\overline{EC} \parallel \overline{AB}$



10. **[Answer]** 1. Try to prove $\triangle BAE \cong \triangle ABC$.

Since $AD = BD$, we have $m\angle DAB = m\angle DBA$.

Since $\overline{AE} \parallel \overline{BD}$, $\overline{BC} \parallel \overline{AD}$, we can get $m\angle CBD = m\angle BDA = m\angle EAD$.

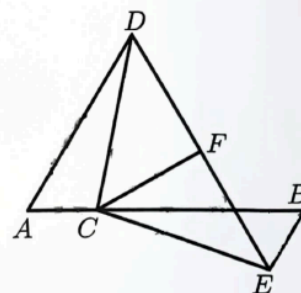
Thus $m\angle CBA = m\angle EAB$.

For triangle ABC and triangle BAE , we have $AB = BA$, $m\angle CBA = m\angle EAB$, $AE = BC$, we can get $\triangle ABC \cong \triangle BAE$.

Thus $AC = BE$.

2. We can get $\triangle BEC \cong \triangle ACE$ by SSS , then suppose $m\angle AEC = m\angle BCE = y$, $m\angle CBA = m\angle EAB = x$, $2x + 2y = 360^\circ$, $x + y = 180^\circ$, then $\overline{EC} \parallel \overline{AB}$.

- 3 As shown in the figure below, point C is on segment AB , $\overline{AD} \parallel \overline{EB}$, $AC = BE$, $AD = BC$, $\overline{CF}$ bisects $\angle DCE$. Prove: $\overline{CF} \perp \overline{DE}$.



3. **【 Answer 】** Shown in the analysis.

【 Solution 】 Since $\overline{AD} \parallel \overline{EB}$,

we have $m\angle A = m\angle B$,

In $\triangle ACD$ and $\triangle BEC$,

$$\begin{cases} AD = BC \\ m\angle A = m\angle B, \\ AC = BE \end{cases}$$

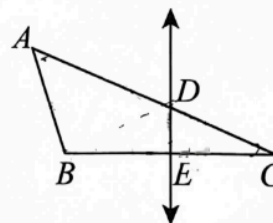
so $\triangle ACD \cong \triangle BEC$ (SAS),

thus $CD = CE$.

Since $\overline{CF}$ bisects $\angle DCE$,

we have $\overline{CF} \perp \overline{DE}$.

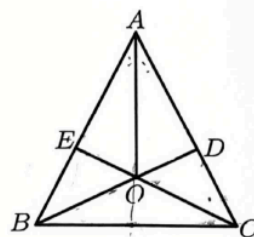
- 6 As shown in the figure below, in $\triangle ABC$, perpendicular bisector of $\overline{BC}$ intersects $\overline{AC}$ and $\overline{BC}$ at D and E , respectively. If $AB = CD$, $m\angle C = 20^\circ$, then $m\angle A = \underline{\hspace{1cm}}$.



6. 【 Answer 】 40°

【 Solution 】 Connect $\overline{DB}$, and we have $DB = DC$,
then $m\angle DBC = m\angle C$,
so $m\angle BDA = 2m\angle C$.
Since $AB = CD$, $DB = DC$,
we have $BA = BD$,
then $m\angle A = m\angle BDA$,
thus $m\angle A = 2m\angle C$.
Since $m\angle C = 20^\circ$,
we have $m\angle A = 40^\circ$.

7. As shown in the figure below, in $\triangle ABC$, $\overline{BD} \perp \overline{AC}$ at D , $\overline{CE} \perp \overline{AB}$ at E . $BD = CE$ and they intersect at O . Connect $\overline{AO}$. Prove: $\overleftrightarrow{AO}$ is the perpendicular bisector of BC .



7. 【 Answer 】 Shown in the analysis.

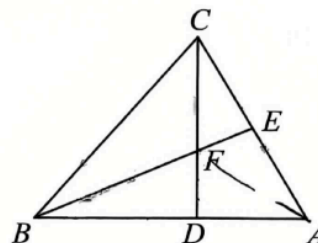
【 Solution 】 Since $\overline{BD} \perp \overline{AC}$ at D , $\overline{CE} \perp \overline{AB}$ at E ,
we have $m\angle BEC = m\angle BDC = 90^\circ$,
In $\text{Rt}\triangle BEC$ and $\text{Rt}\triangle CDB$,
$$\begin{cases} BC = CB \\ CE = BD \end{cases}$$

Therefore $\triangle BEC \cong \triangle CDB$,
then $m\angle ABC = m\angle ACB$, $m\angle ECB = m\angle DBC$,
thus $AB = AC$, $BO = OC$,
so Point A and O are on the perpendicular bisector of $\overline{BC}$,
finally $\overleftrightarrow{AO}$ is the perpendicular bisector of $\overline{BC}$.

- 8 As shown in the figure below, in $\triangle CBD$, $CD = BD$, $\overline{CD} \perp \overline{BD}$, $\overline{BE}$ bisects $\angle CBA$ and intersects $\overline{CD}$ at F . $\overline{CE} \perp \overline{BE}$ at E , $\overline{CE}$ intersects $\overline{BD}$ at A .

(1) Prove: $BF = AC$.

(2) $\overleftrightarrow{BE}$ is the perpendicular bisector of $\overline{AC}$.



8. 【 Answer 】 (1) Shown in the analysis.

(2) Shown in the analysis.

(3) $2\sqrt{2} + 4$

【 Solution 】 (1) Since $\overline{BE} \perp \overline{AC}$,

we have $m\angle ABE + m\angle A = 90^\circ$.

Since $\overline{CD} \perp \overline{AB}$, $m\angle ADC = m\angle BDF = 90^\circ$

then $m\angle A + m\angle ACD = 90^\circ$,

thus $m\angle ABE = m\angle ACD$,

and $BD = CD$,

we have $\triangle BDF \cong \triangle CDA$ (ASA),

so $BF = AC$.

(2) $\overline{BE}$ bisects $\angle ABC$, $\therefore m\angle ABE = m\angle CBE$,

$m\angle CBE = m\angle ACD$, $m\angle ADC = m\angle BEC = 90^\circ$,

so $m\angle A = m\angle BCA$, $\triangle ABC$ is an isosceles triangle, $AB = BC$.

Since $\overline{BE} \perp \overline{AC}$,

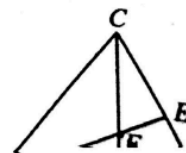
we have $\overleftrightarrow{BE}$ is the perpendicular bisector of $\overline{AC}$.

(3) Connect AF , $AD = DF = 2$, $m\angle AFD = m\angle CBD = 45^\circ$,

so $AF = 2\sqrt{2}$, $m\angle FAC = m\angle ACF$,

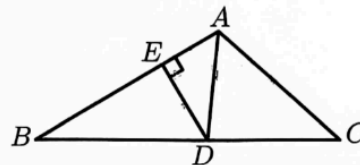
then $CF = AF = 2\sqrt{2}$

thus $BD = CD = 2 + 2\sqrt{2}$



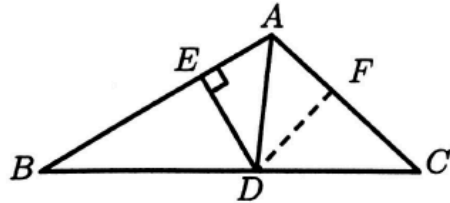
L6 E

- 4 As shown in the figure, $\overline{AD}$ is the bisector of $\angle BAC$ in $\triangle ABC$. If $\overline{DE} \perp \overline{AB}$ at point E , $DE = 2$ and $AC = 3$, then the area of $\triangle ADC$ is _____.



4. **【 Answer 】** 3

【 Solution 】 Draw $\overline{DF} \perp \overline{AC}$ at F ,

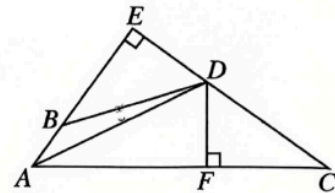


since $\overline{AD}$ is the bisector of $\angle BAC$ in $\triangle ABC$, $\overline{DE} \perp \overline{AB}$ and $\overline{DF} \perp \overline{AC}$,
we have $DF = DE = 2$,

so the area of $\triangle ADC$ is $= \frac{1}{2} \times AC \times DF = 3$.

So the answer is 3.

6. As shown in the figure, $BE = CF$, $\overline{DE} \perp \overline{AB}$ and intersects line AB at point E . $\overline{DF} \perp \overline{AC}$ at point F and $DB = DC$. Prove that $\overline{AD}$ is the bisector of $\angle EAC$.



6. **【 Answer 】** See analysis.

【 Solution 】 Paragraph proof:

Since $\overline{DE} \perp \overline{AB}$ at E and $\overline{DF} \perp \overline{AC}$ at point F ,

we have $m\angle BED = m\angle CFD = 90^\circ$,

and therefore $\triangle BDE$ and $\triangle CDF$ are right triangles.

In $\text{Rt}\triangle BDE$ and $\text{Rt}\triangle CDF$,

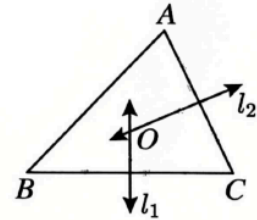
we have $\begin{cases} BE = CF \\ BD = CD \end{cases}$,

and therefore $\text{Rt}\triangle BDE \cong \text{Rt}\triangle CDF$ (HL).

So $DE = DF$,

and therefore $\overline{AD}$ is the bisector of $\angle BAC$.

10. As shown in the figure below, l_1 is the perpendicular bisector of $\overline{BC}$ and l_2 is the perpendicular bisector of $\overline{AC}$. l_1 and l_2 intersect at O . Prove: Point O is on the perpendicular bisector of $\overline{AB}$.



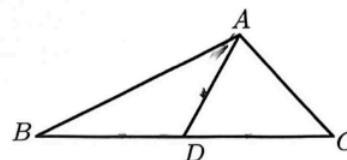
10. **【 Answer 】** Shown in the analysis.

【 Solution 】 Plan for proof:

Connect $\overline{OA}$, $\overline{OB}$ and $\overline{OC}$. Since two perpendicular lines intersect at O , then we have $OA = OB = OC$. From $OA = OB$, we can prove that O is also on the perpendicular bisector of $\overline{AB}$.

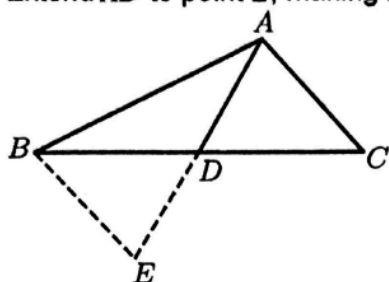
L7

5. In $\triangle ABC$, $BD = CD$, $\angle DAC = \angle ABC + \angle ACB$. Given that $AD = 3$, find the length of $AB = \underline{\hspace{2cm}}$.



5. **【 Answer 】** 6

【 Solution 】 Extend AD to point E , making $DE = AD$, connect BE ,



$$\text{In } \triangle ADC \text{ and } \triangle EDB, \begin{cases} AD = ED \\ \angle ADC = \angle EDB, \\ CD = BD \end{cases}$$

$\therefore \triangle ADC \cong \triangle EDB$ (SAS),

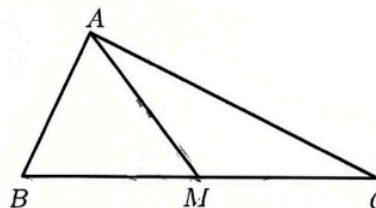
$\therefore \angle DAC = \angle E$, $\angle C = \angle EBD$,

$\therefore \angle ABE = \angle ABC + \angle EBD = \angle ABC + \angle C$, $\angle DAC = \angle B + \angle C$,

$\therefore \angle ABE = \angle DAC = \angle E$,

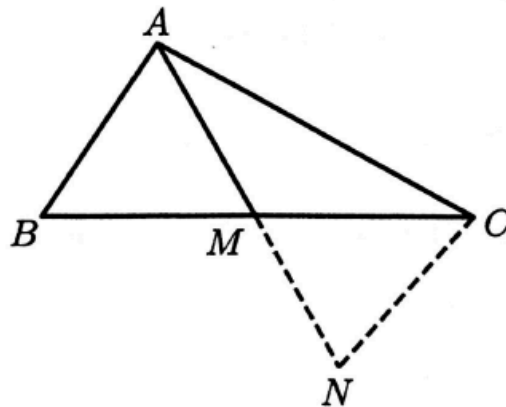
$\therefore AB = AE = 2AD = 6$.

6. As shown in the figure, in $\triangle ABC$, $\overline{AM}$ is the median. Prove that $AM < \frac{1}{2}(AB + AC)$.



6. **【 Answer 】** See analysis.

【 Solution 】 Extend $\overline{AM}$ to point N such that $AM = MN$.



In $\triangle ABM$ and $\triangle NCM$,

$$\begin{cases} AM = MN \\ BM = CM \\ m\angle AMB = m\angle NMC \end{cases},$$

we have $\triangle ABM \cong \triangle NCM$ (SAS),

and so $AB = NC$.

In $\triangle ACN$,

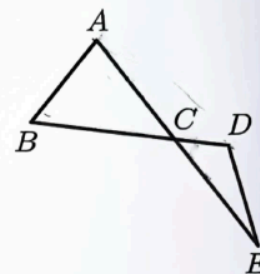
$$AC + NC > AN,$$

which means $AC + AB > 2AM$,

$$AM < \frac{1}{2}(AB + AC).$$

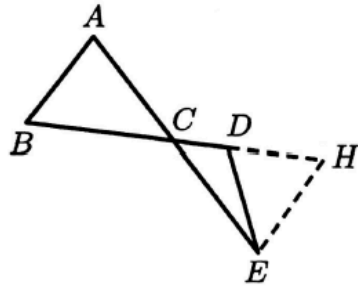
L7 E

26 As shown in the figure below, $m\angle B + m\angle CDE = 180^\circ$, $AC = CE$. Prove: $AB = DE$.



2. **【 Answer 】** Shown in the analysis.

【 Solution 】 Extend $\overline{CD}$ to H so that $CH = BC$, connect $\overline{EH}$.



In $\triangle ABC$ and $\triangle EHC$

$$\begin{cases} AC = CE \\ m\angle ACB = m\angle ECH, \\ BC = CH \end{cases}$$

so $\triangle ABC \cong \triangle EHC$ (SAS),

we have $AB = EH$, $m\angle B = m\angle H$.

Since $m\angle B + m\angle CDE = 180^\circ$, $m\angle HDE + m\angle CDE = 180^\circ$,

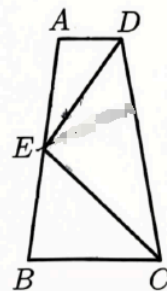
we have $m\angle HDE = m\angle B = m\angle H$,

thus $DE = HE$.

Since $AB = HE$,

we have $AB = DE$.

4. As shown in the figure, in trapezoid $ABCD$, $\overline{AD} \parallel \overline{BC}$, E is the midpoint of $\overline{AB}$, $\overline{DE} \perp \overline{CE}$, $AD = 3$, and $CD = 9$. Then $BC = \underline{\hspace{2cm}}$.



4. 【 Answer 】 6

【 Solution 】 Extend $\overline{DE}$ and $\overline{CB}$ such that they intersect at point F .

Since $\overline{AD} \parallel \overline{BC}$,

we have $m\angle A = m\angle ABF$ and $m\angle ADE = m\angle F$.

It is easy to show that $\triangle AED \cong \triangle BEF$,

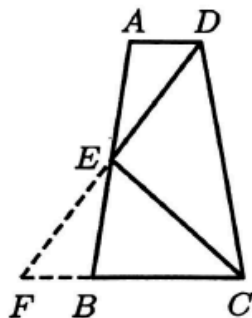
so we have $AD = BF$ and $DE = EF$.

Since $\overline{DE} \perp \overline{CE}$,

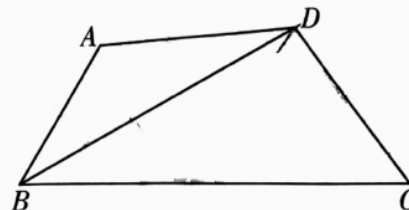
we have $CD = CF = BC + BF$.

Therefore $AD + BC = DC$,

so $BC = 9 - 3 = 6$.

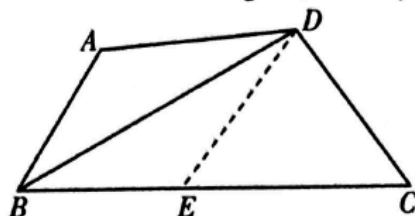


6 As shown in the figure, given that in quadrilateral $ABCD$, $BC > AB$, $AD = CD$, $\overline{BD}$ bisects $\angle ABC$, prove that $m\angle A + m\angle C = 180^\circ$.



6. **【 Answer 】** See analysis.

【 Solution 】 As shown in the figure, find a point E such that $BE = BA$, draw $\overline{DE}$,



Since $\overline{BD}$ bisects $\angle ABC$,
we have $m\angle ABD = m\angle EBD$.

In $\triangle BAD$ and $\triangle BED$,

since $\begin{cases} BA = BE \\ m\angle ABD = m\angle EBD, \\ BD = BD \end{cases}$

we have $\triangle BAD \cong \triangle BED$ (SAS),

so $DA = DE$ and $m\angle A = m\angle BED$.

Since $DC = AD$,

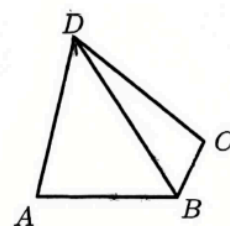
we have $DE = DC$.

Therefore $m\angle C = m\angle DEC$.

Since $m\angle BED + m\angle DEC = 180^\circ$,

we have $m\angle A + m\angle C = 180^\circ$.

7. As shown in the figure, in $ABCD$, $\overline{BD}$ bisects $\angle ABC$ and $m\angle A + m\angle C = 180^\circ$. If $AB > BC$, prove that $AD = DC$.



7. **【 Answer 】** See analysis.

【 Solution 】 Take a point E from $\overline{AB}$ such that $BE = BC$.

Draw $\overline{DE}$, then $\triangle DCB \cong \triangle DEB$,

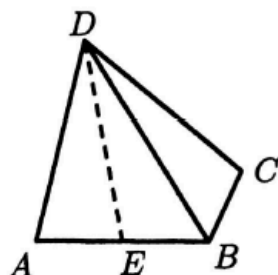
so $m\angle DCB = m\angle DEB$, $DC = DE$.

Since $m\angle A + m\angle C = 180^\circ$, we have $m\angle DEA + m\angle DEB = 180^\circ$,

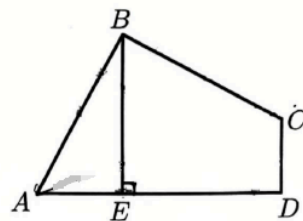
so $m\angle A = m\angle DEA$,

and therefore $DA = DE$.

As such, $AD = DC$.

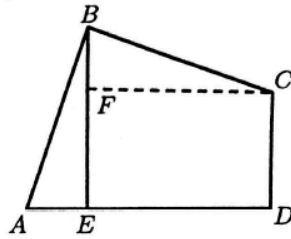


9 In the figure, in quadrilateral $ABCD$, $AB = BC$, $m\angle ABC = m\angle CDA = 90^\circ$, $\overline{BE} \perp \overline{AD}$ at E , and the area of quadrilateral $ABCD$ is 10. The length of $\overline{BE}$ is _____.



9. [Answer] $\sqrt{10}$

[Solution]



As shown in the figure, let $\overline{CF} \perp \overline{BE}$ at point F .

Since $m\angle ABC = 90^\circ$, we have $m\angle BAE + m\angle ABE = m\angle ABE + m\angle CBF = 90^\circ$.

In $\triangle AEB$ and $\triangle BFC$, we have:
$$\begin{cases} m\angle BAE = m\angle CBF \\ m\angle AEB = m\angle BFC \\ AB = BC \end{cases}$$

Therefore, $\triangle AEB$ is congruent to $\triangle BFC$ (AAS).

Thus, $AE = BF$ and $BE = CF$.

Let $AE = BF = a$ and $BE = CF = b$.

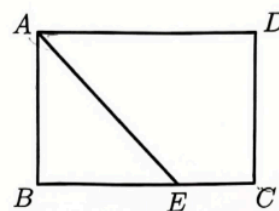
Since the area of quadrilateral $ABCD$ is 10, we have $\frac{1}{2}ab + \frac{1}{2}ab + (b-a)b = 10$.

Simplifying, we get $b^2 = 10$.

Therefore, $b = \sqrt{10}$.

Thus, $BE = \sqrt{10}$.

- 5 In rectangle $ABCD$, $AB = 3$, $BC = 4$, and E is a point on $\overline{BC}$ which is not at point B or point C . If AE is an integer, $AE = \underline{\hspace{1cm}}$.



A. 3

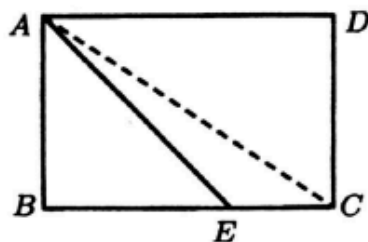
B. 4

C. 5

D. 6

5. **【 Answer 】** B

【 Solution 】 Draw $\overline{AC}$.



Since $ABCD$ is a rectangle,

$$m\angle B = 90^\circ.$$

In $\text{Rt}\triangle ABC$, $AB = 3$, and $BC = 4$,

$$\text{so } AC = \sqrt{AB^2 + BC^2} = 5.$$

Therefore $AB < AE < AC$,

$$\text{so } 3 < AE < 5.$$

Since AE is an integer,

$$AE = 4.$$

Therefore, the correct choice is B.

- 8 In triangle ABC , $AB = AC$, $AC = 2$, and $m\angle BAC = 30^\circ$. Find the length of $\overline{BC}$.

8. 【 Answer 】 $\sqrt{6} - \sqrt{2}$

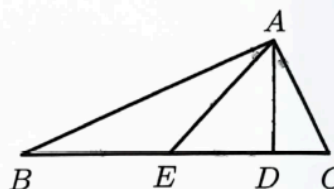
【 Solution 】 Add $\overline{BD}$ through B such that $\overline{BD} \perp \overline{AC}$ at D .

$$BD = 1, AD = \sqrt{3}, CD = 2 - \sqrt{3}.$$

$$BC^2 = 1^2 + (2 - \sqrt{3})^2 = 8 - 4\sqrt{3}, BC = \sqrt{6} - \sqrt{2}.$$

L8 E

- 7 As shown in the figure, in $\triangle ABC$, $m\angle BAC = 90^\circ$, $\overline{AD} \perp \overline{BC}$ at point D . E is the midpoint of $\overline{BC}$, and $AD = ED = 3$. The length of $\overline{BC}$ is _____.



A. $6\sqrt{2}$

B. $4\sqrt{3}$

C. $3\sqrt{2}$

D. $3\sqrt{2} + 2\sqrt{3}$

1. 【 Answer 】 A

【 Solution 】 Since $\overline{AD} \perp \overline{BC}$,

$$m\angle ADE = 90^\circ.$$

$$\text{Since } AD = ED = 3,$$

$$AE = \sqrt{AD^2 + ED^2} = 3\sqrt{2}.$$

Since $\angle BAC = 90^\circ$, and E is the midpoint of $\overline{BC}$,

$$BC = 2AE = 6\sqrt{2}.$$

Therefore, the correct answer is A.

- 2 Right triangle ABC has leg lengths $AB = 20$ and $BC = 21$. Including $\overline{AB}$ and $\overline{BC}$, how many line segments with integer length can be drawn from vertex B to a point on hypotenuse $\overline{AC}$? (2018 AMC 10A Problem, Question #16)

A. 5

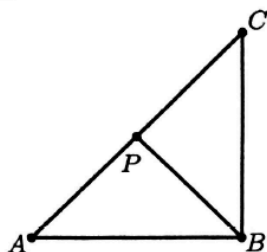
B. 8

C. 12

D. 13

E. 15

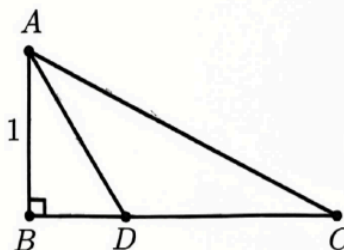
2. **[Answer]** D
[Solution]



As the problem has no diagram, we draw a diagram. The hypotenuse has length 29. Let P be the foot of the altitude from B to $\overline{AC}$. Note that $\overline{BP}$ is the shortest possible length of any segment. Writing the area of the triangle in two ways, we can solve for $BP = \frac{20 \cdot 21}{29}$, which is between 14 and 15.

Let the line segment be $\overline{BX}$, with X on $\overline{AC}$. As you move X along the hypotenuse from A to P , the length of $\overline{BX}$ strictly decreases, hitting all the integer values from 20, 19, ..., 15 (IVT). Similarly, moving X from P to C hits all the integer values from 15, 16, ..., 21. This is a total of **(D) 13** line segments.

4. Triangle ABC has a right angle at B , $AB = 1$, and $BC = 2$. The bisector of $\angle BAC$ meets $\overline{BC}$ at D . What is BD ? (2009 AMC 10B Problem, Question #20)



- A. $\frac{\sqrt{3}-1}{2}$ B. $\frac{\sqrt{5}-1}{2}$ C. $\frac{\sqrt{5}+1}{2}$ D. $\frac{\sqrt{6}+\sqrt{2}}{2}$ E. $2\sqrt{3}-1$

4. **[Answer]** B

[Solution] By the Pythagorean Theorem, $AC = \sqrt{5}$. Then, from the Angle Bisector Theorem, we have:

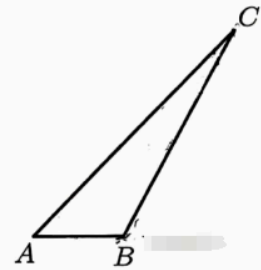
$$\frac{BD}{1} = \frac{2-BD}{\sqrt{5}}$$

$$BD \left(1 + \frac{1}{\sqrt{5}} \right) = \frac{2}{\sqrt{5}}$$

$$BD(\sqrt{5}+1) = 2$$

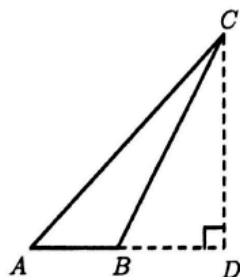
$$BD = \frac{2}{\sqrt{5}+1} = \boxed{\frac{\sqrt{5}-1}{2}} \Rightarrow B.$$

- 10 As shown in the figure, in $\triangle ABC$, $m\angle A = 45^\circ$, $m\angle C = 15^\circ$, and $AB = 10$. Calculate the area of $\triangle ABC$.



10. **[Answer]** $25\sqrt{3} + 75$

[Solution]



Extend $\overline{AB}$ and draw $\overline{CD} \perp \overline{AD}$ through C such that they intersect at point D .

Since $\overline{CD} \perp \overline{AD}$,

$m\angle ADC = 90^\circ$,

so $m\angle A + m\angle ACD = 90^\circ$.

Since $m\angle A = 45^\circ$,

$m\angle A = m\angle ACD = 45^\circ$,

so $AD = CD$.

Let $BD = x$;

since $AD = 10 + x$,

we have $CD = 10 + x$.

Since $m\angle ACB = 15^\circ$,

$m\angle BCD = m\angle ACD - m\angle ACB = 30^\circ$,

so in $\text{Rt}\triangle BCD$, $BC = 2BD = 2x$,

$BC^2 = BD^2 + CD^2$.

Therefore $CD = \sqrt{BC^2 - BD^2} = \sqrt{4x^2 - x^2} = \sqrt{3}x$,

so $10 + x = \sqrt{3}x$,

so $x = \frac{10}{\sqrt{3} - 1} = 5\sqrt{3} + 5$,

so $CD = 10 + x = 5\sqrt{3} + 15$.

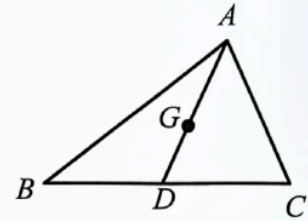
$S_{\triangle ABC} = \frac{1}{2} \times AB \times CD$

$= \frac{1}{2} \times 10 \times (5\sqrt{3} + 15)$

$= 25\sqrt{3} + 75$.

L9

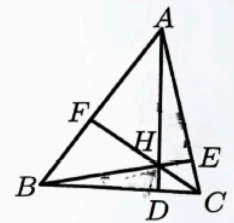
2. As shown in the figure below, in $\triangle ABC$, $\overline{AD}$ is the median from A to $\overline{BC}$ and G is the centroid. If $AG = 6$, then the length of $\overline{DG}$ is _____.



- A. 1 B. 2 C. 3 D. 6

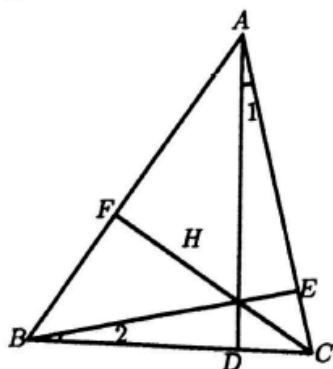
Ans C

7. In the figure, H is the orthocenter (the intersection point of the three altitudes) of acute triangle ABC . If $AH = BC$, then the degree measure of angle BAC is _____.



7. **[Answer]** 45°

[Solution]



Since H is the orthocenter of acute triangle ABC , we have $\overline{CF} \perp \overline{AB}$, $\overline{BE} \perp \overline{AC}$, and $\overline{AD} \perp \overline{BC}$.

Therefore, $m\angle AEH = m\angle BEC = 90^\circ$.

Thus, $m\angle DAC + m\angle ACB = m\angle EBC + m\angle ACB = 90^\circ$.

This implies that $m\angle 1 + m\angle ACD = m\angle 2 + m\angle ACD$.

Therefore, $m\angle 1 = m\angle 2$.

Also, since $AH = BC$,

we have $\triangle AEH \cong \triangle BEC$ (AAS).

Thus, $AE = BE$.

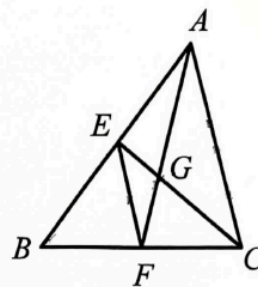
Therefore, $\triangle ABE$ is an isosceles right triangle.

Hence, $m\angle BAE = 45^\circ$, which means $m\angle BAC = 45^\circ$.

Therefore, the answer is 45° .

L9 E

- 2 In the figure, point G is the centroid of triangle $\triangle ABC$. Connect $\overline{CG}$ and $\overline{AG}$, and extend them to intersect $\overline{AB}$ and $\overline{BC}$ at points E and F . Connect $\overline{EF}$. If $AC = 3.4$, the length of $\overline{EF}$ is _____.



A. 1.7

B. 1.8

C. 2.2

D. 3.4

2. **【 Answer 】** A

【 Solution 】 Since point G is the centroid of triangle $\triangle ABC$, point E is the midpoint of $\overline{AB}$, and point F is the midpoint of $\overline{BC}$, therefore $\overline{EF}$ is the median of triangle $\triangle ABC$. Hence, $EF = \frac{1}{2}AC$.

Since $AC = 3.4$, we have $EF = 1.7$. Therefore, the correct answer is A.

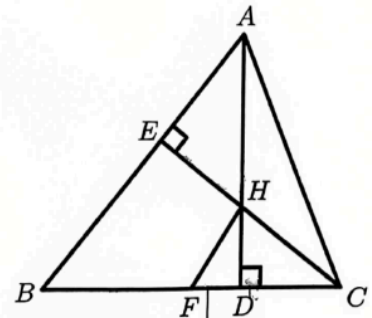
3. The lengths of the three sides of the triangle are 10, 10, and $10\sqrt{3}$. Find the distance from the incenter of a triangle to its longest side.

【 Answer 】 $10\sqrt{3} - 15$

【 Solution 】 N/A

5. In the acute triangle ABC , let $\overline{AD}$ and $\overline{CE}$ be the altitudes, and let H be the orthocenter. Given that $AD = BC$ and F is the midpoint of $\overline{BC}$, prove that:

$$FH + DH = \frac{1}{2}BC.$$



5. 【 Answer 】 See analysis.

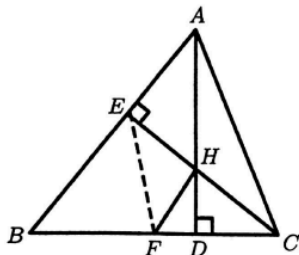
【 Solution 】 As shown in the figure, connect $\overline{EF}$.

Then $EF = CF = \frac{1}{2}BC$. Therefore, $FH^2 = EF^2 - EH \cdot CH = EF^2 - AH \cdot HD$.

$$= EF^2 - AH \cdot HD - HD^2 + HD^2 = EF^2 - HD \cdot AD + HD^2$$

$$= EF^2 - HD \cdot BC + HD^2 = EF^2 - 2HD \cdot EF + HD^2 = (EF - HD)^2.$$

Since $EF > FH > HD$, we have $FH = EF - DH = \frac{1}{2}BC - DH$.



6 In the isosceles triangle $\triangle ABC$, $AB = AC = 6$, $BC = 6\sqrt{3}$, and I is the incenter of $\triangle ABC$.

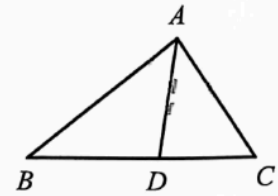
(1) Find $m\angle BIC$.

(2) Find AI .

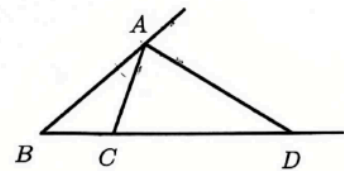
6. 【 Answer 】 (1) 150°
(2) $12 - 6\sqrt{3}$

8 Use parallel line to prove the following:

(1) As shown in the figure below, $\overline{AD}$ is the angle bisector of $\triangle ABC$, prove that $\frac{AB}{AC} = \frac{BD}{CD}$.



(2) In the figure, the external angle bisector of $\angle BAC$ intersects the extension of side BC at point D . Prove that $\frac{AB}{AC} = \frac{BD}{CD}$.



8. 【Answer】 (1) See analysis.

(2) See analysis.

【Solution】 (1) Construct "A" model:

Passing through point C , draw a line parallel to $\overline{AD}$ and intersect the extension of $\overline{BA}$ at point E .

$$\text{Then } \frac{AB}{AE} = \frac{BD}{DC}, (1).$$

Since $\overline{CE} \parallel \overline{AD}$,

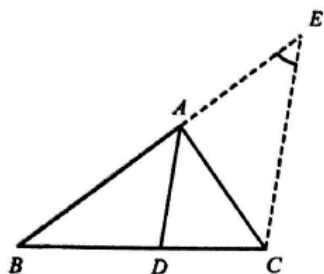
$$m\angle DAC = m\angle ACE, m\angle BAD = m\angle AEC,$$

From the given information, we know that $m\angle BAD = m\angle DAC$,

therefore, $m\angle ACE = m\angle AEC$,

thus, $AC = AE$, (2)

substituting (2) into (1), we have $\frac{AB}{AC} = \frac{BD}{DC}$.



Construct "A" model:

Through point D , draw a line parallel to $\overline{AC}$ intersecting $\overline{AB}$ at point E ,

$$\text{then } \frac{BE}{AE} = \frac{BD}{DC}.$$

Since $\overline{DE}$ is parallel to $\overline{AC}$,

we have $m\angle EDA = m\angle DAC$.

From the given information, we know that $m\angle EAD = m\angle DAC$.

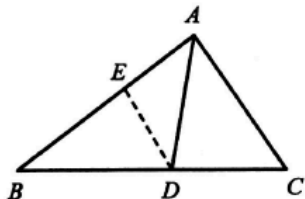
Therefore, $m\angle EAD = m\angle EDA$.

Thus, $AE = DE$, so $\frac{BE}{DE} = \frac{BD}{DC}$ (1).

Also, since $\overline{DE}$ is parallel to $\overline{AC}$, we have $\triangle BED \sim \triangle BAC$.

Therefore, $\frac{BE}{AB} = \frac{DE}{AC}$, which implies $\frac{BE}{DE} = \frac{AB}{AC}$ (2).

From (1) and (2), we conclude that $\frac{AB}{AC} = \frac{BD}{DC}$.



Construct "8" model:

Through point D , draw a line parallel to $\overline{AB}$ intersecting the extension of $\overline{AD}$ at point E .

Then, we have $\triangle ABD \sim \triangle ECD$.

Therefore, we have $\frac{AB}{CE} = \frac{BD}{DC}$, (1).

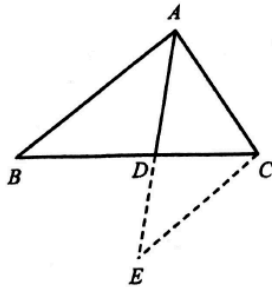
Also, since $\overline{CE} \parallel \overline{AB}$, we have $m\angle BAD = m\angle AEC$.

From the given information, we know that $m\angle BAD = m\angle EAC$.

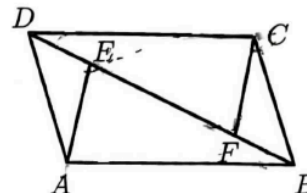
Therefore, $m\angle EAC = m\angle AEC$.

Thus, $AC = CE$, (2).

From (1) and (2), we can conclude that $\frac{AB}{AC} = \frac{BD}{DC}$.



- As shown in the figure, in parallelogram $ABCD$, points E and F are on $\overline{BD}$, and $DF = BE$. Hypothesize the positional and quantitative relationships between $\overline{AE}$ and $\overline{CF}$ and prove them.



5. **【 Answer 】** See analysis

【 Solution 】 Hypothesis: $AE = CF$ and $\overline{AE} \parallel \overline{CF}$.

Proof: since $ABCD$ is a parallelogram,
we have $\overline{AB} \parallel \overline{CD}$ and $AB = CD$,
so $m\angle ABE = m\angle CDF$.

In $\triangle ABE$ and $\triangle CDF$,

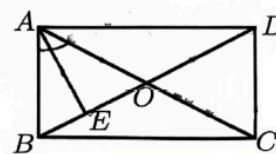
$$\begin{cases} AB = CD \\ m\angle ABE = m\angle CDF, \\ BE = DF \end{cases}$$

so $\triangle ABE \cong \triangle CDF$ (SAS).

So $AE = CF$ and $m\angle AEB = m\angle CFD$.

Therefore $\overline{AE} \parallel \overline{CF}$.

- 8 As shown in the figure, in rectangle $ABCD$, diagonals $\overline{AC}$ and $\overline{BD}$ intersect at point O . Draw $\overline{AE} \perp \overline{BD}$ through A such that the perpendicular foot is located at E . If $m\angle OCD = 56^\circ$, then $m\angle EAO = \underline{\hspace{1cm}}^\circ$.



8. 【 Answer 】 22

【 Solution 】 Since $\overline{AE} \perp \overline{BD}$,

we have $m\angle AEB = m\angle AED = 90^\circ$.

Since $m\angle OCD = 56^\circ$ and that $ABCD$ is a rectangle,

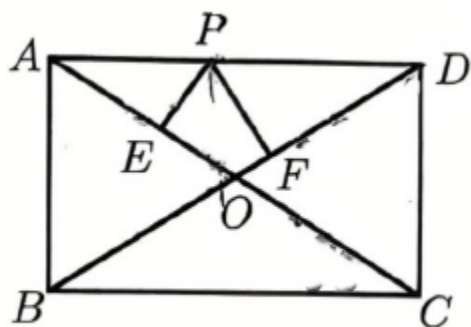
we have $AC = BD$ and $OC = OD$,

so $m\angle OCD = m\angle ODC = 56^\circ$.

Therefore $m\angle DOC = 180^\circ - 2m\angle OCD = 68^\circ$, and $m\angle DOC = m\angle AOB$.

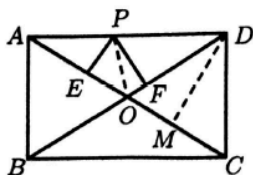
As such, $m\angle EAO = 180^\circ - m\angle AEO - m\angle AOB = 180^\circ - 90^\circ - 68^\circ = 22^\circ$.

9. As shown in the figure, in rectangle $ABCD$, $AB = 3$, $AD = 4$, and P is a moving point on $\overline{AD}$. Given that $\overline{PF} \perp \overline{BD}$ at point F and that $\overline{PE} \perp \overline{AC}$ at point E , then $PE + PF$ is _____.



9. 【 Answer 】 $\frac{12}{5}$ Alternative 2 $\frac{2}{5}$ Alternative 2.4

【 Solution 】 Draw $\overline{OP}$, and draw $\overline{DM} \perp \overline{AC}$ through D at M , as shown below:



Since $ABCD$ is a rectangle,

we have $AO = OC = \frac{1}{2}AC$, $OD = OB = \frac{1}{2}BD$, $AC = BD$, and $m\angle ADC = 90^\circ$.

So $OA = OD$.

Then by pythagorean theorem, $AC = \sqrt{3^2 + 4^2} = 5$.

Since $S_{\triangle ADC} = \frac{1}{2} \times 3 \times 4 = \frac{1}{2} \times 5 \times DM$,

we have $DM = \frac{12}{5}$.

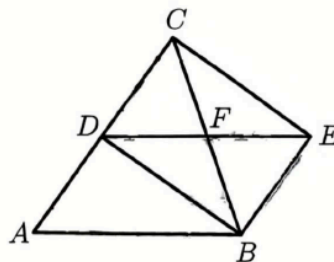
Since $S_{\triangle AOD} = S_{\triangle APO} + S_{\triangle DPO}$,

we have $\frac{1}{2}(AO \times DM) = \frac{1}{2}(AO \times PE) + \frac{1}{2}(DO \times PF)$.

This means that $PE + PF = DM = \frac{12}{5}$.

Therefore the correct answer is B.

- 10 As shown in the figure, in $\triangle ABC$, $AB = BC$, $\overline{BD}$ bisects $\angle ABC$. Draw a parallel line segment with respect to $\overline{AB}$ through point D ; draw a parallel line segment with respect to $\overline{AC}$ through B , such that two line segments intersect at point E and that $\overline{BC}$ intersect with $\overline{DE}$ at point F . Draw $\overline{CE}$. Prove that quadrilateral $BECD$ is a rectangle.



10. **【 Answer 】** See analysis

【 Solution 】 Since $AB = BC$, and $\overline{BD}$ bisects $\angle ABC$, we have $AD = DC$ and $\overline{BD} \perp \overline{CA}$.
 Since $\overline{AB} \parallel \overline{DE}$ and $\overline{AD} \parallel \overline{BE}$, we have $ABED$ being a parallelogram.
 Therefore $AD = BE$, $\overline{AD} \parallel \overline{BE}$, and $AB = DE$.
 So $DC = BE$ and $\overline{DC} \parallel \overline{BE}$.
 Therefore $BECD$ is also a parallelogram.
 Since $\overline{BD} \perp \overline{CA}$, we have $m\angle BDC = 90^\circ$.
 So quadrilateral $BECD$ is a rectangle.

L10 E

- 4 With respect to quadrilateral $ABCD$, six conditions were given:

- ① $\overline{AB} \parallel \overline{CD}$; ② $\overline{BC} \parallel \overline{AD}$;
 ③ $AB = CD$; ④ $BC = AD$;
 ⑤ $m\angle A = m\angle C$; ⑥ $m\angle B = m\angle D$.

The number of ways to prove that $ABCD$ is a parallelogram by choosing two of the above conditions is _____.

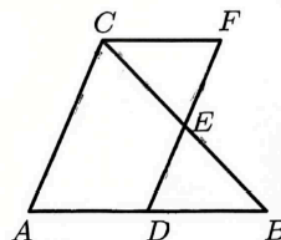
- A. 4 B. 9 C. 13 D. 15

4. **【 Answer 】** B

【 Solution 】 All the situations (a total of nine):

①②, ①③, ①⑤, ①⑥, ②④, ②⑤, ②⑥, ③④, ⑤⑥.

- 5 As shown in the figure, in $\triangle ABC$, D and E are the midpoints of $\overline{AB}$ and $\overline{BC}$, respectively, and the point F is on the extension of $\overline{DE}$. If a condition is added to make the quadrilateral $ADFC$ a parallelogram, then this condition is _____.



- A. $m\angle B = m\angle F$ B. $m\angle B = m\angle BCF$ C. $AC = CF$

D. $AD = CF$

5. **【 Answer 】** B

【 Solution 】 Since in $\triangle ABC$, D and E are midpoints of $\overline{AB}$ and $\overline{BC}$, $\overline{DE}$ is the midsegment of $\triangle ABC$.

So $DE = \frac{1}{2}AC$ and $\overline{AD} \parallel \overline{FC}$.

A: by $m\angle B = m\angle F$, it is impossible to deduce that $\overline{DE} \parallel \overline{AC}$, so it is impossible to prove that $ADFC$ is a parallelogram. So A is incorrect.

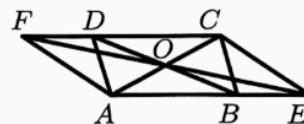
B: by $m\angle B = m\angle BCF$, we have $\overline{CF} \parallel \overline{AB}$, which means that $\overline{AD} \parallel \overline{FC}$.

Then prove that $ADFC$ is a parallelogram through the theorem "a quadrilateral with two pairs of opposite sides parallel is a parallelogram". So B is correct.

C: by $AC = CF$, it is impossible to prove that $\overline{AD} \parallel \overline{DF}$. So it is impossible to prove that $ADFC$ is a parallelogram. So C is incorrect.

D: by $AD = CF$, it is impossible to prove that $ADFC$ is a parallelogram, so D is incorrect. Therefore, the correct answer is B.

- 6 As shown in the figure, the diagonals of the parallelogram $ABCD$ intersect at point O , and the line EF passes through the point O , intersecting with the extensions of $\overline{AB}$ and $\overline{CD}$ at the point E and F , respectively. Prove: The quadrilateral $AECF$ is a parallelogram.



6. **【 Answer 】** See analysis

【 Solution 】 $ABCD$ is given to be a parallelogram,

so $OD = OB$ and $OA = OC$.

Since $\overline{AB} \parallel \overline{CD}$,

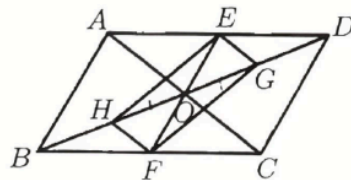
we have $m\angle DFO = m\angle BEO$ and $m\angle FDO = m\angle EBO$.

From here it is easy to prove that $\triangle FDO \cong \triangle EBO$ (AAS),

so $OF = OE$.

Therefore $AECF$ is indeed a parallelogram .

- As shown in the figure, draw a line EF to meet $\overline{AD}$ and $\overline{BC}$ at E and F , respectively through O , the intersection point of the diagonals of parallelogram $ABCD$. It is given that G and H are the midpoints of $\overline{OD}$ and $\overline{OB}$ respectively. Prove that quadrilateral $EHFG$ is a parallelogram.



7. **【 Answer 】** See analysis.

【 Solution 】 It is easy to see that $\triangle AOE \cong \triangle COF$.

So $EO = FO$.

Since $OB = OD$, $OH = \frac{1}{2}OB$, and $OG = \frac{1}{2}OD$,

we have $OH = OG$.

So $EHFG$ is a parallelogram.